3D VIDEO TRACKING TECHNOLOGY IN THE ASSESSMENT OF OROFACIAL IMPAIRMENTS IN NEUROLOGICAL DISORDERS

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Abstract & Keywords

Changes to the orofacial muscles' movement and speech are often among the earliest signs perceived in Neurological disorders. Detection of subtle changes in speech and facial movements can help with the diagnosis and prognosis of neurological disorders. Deep artificial intelligent video-based facial analysis models have the potential to be used as objective and non-invasive clinical tools. This thesis used the V3 framework for evaluation of digital biomarkers and their adoption into clinical settings to evaluate an automatic video-based facial analysis system as an objective assessment tool for accessing orofacial movements. The proposed system consists of a 3D camera and Artificial Intelligent-based algorithms that automatically extract objective clinically interpretable kinematic features from video recordings of individuals performing standard orofacial tasks. This work investigates the analytical and clinical validation of the proposed system to assess the severity of orofacial impairment in clinical groups.

Keywords: computer-aided orofacial assessment, neurological disorders, facial kinematics

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List of Abbreviations, symbols, and nomenclatures

2D – 2 Dimensional
3D – 3 Dimensional
AI – Artificial Intelligence
ALS – Amyotrophic Lateral Sclerosis
FAN – Facial Alignment Network
HC – Healthy Control
LSTM – Long Short-Term Memory
MS – Multiple Sclerosis
MND – Motor Neuron Disease
NME – Normalized Mean Error
PD – Parkinson Disease
RMSE – Root Mean Square Error

Chapter 1 – Introduction

Neurological disorders

There is a rapid change in the demographic of the world as life expectancy is being prolonged due to several factors such as better healthcare, improved living conditions, and technology advancements [1]. In fact, it is estimated that the older adults aged 60 and over will account for an estimated 22% of global population by year 2050 [2]. Despite the above-mentioned improvements in the life expectancy, neurological conditions have arguably become the most dreaded condition of older adults, with age being the main risk factor for such disorders, including Alzheimer disease (AD), Parkinson disease (PD), motor neuron disease (MND), and stroke [3]. Treating and caring for individuals with such conditions are costly and expected to multiply in the years to come. It is estimated that caring for older adults with neurological conditions already consumed over $600 billion worldwide [4]. As the global population of older adults is on the rise, so is the prevalence of age-related neurological conditions and the need for development of cheap, efficient, scalable tools that are sensitive to the early signs and subtle changes associated with the onset of neurological disorders.

Neurological disorders are brain, cognitive, or behavioural disorders that may affect various human functions such as speaking, limb movements and cognitive/ learning capacity. There are over six hundred known neurological impairments with the global prevalence of these diseases found to be at 10.2% [5].

The most common observed impairments in neurological conditions are impaired motor, sensory, visual, balance, cognition, swallowing function and communication. Motor impairments include weakness or complete paralysis across various muscle groups e.g., in the muscles of upper or lower limbs, trunk muscles as well as muscles of the head and neck, including orofacial muscles. Sensory impairment includes numbness, loss of sensation of touch, pressure, or pain the affected body structures. Cognitive impairment includes, as example, deficits in attention, memory, or visuo-spatial orientation. A common sign of neurological disorders is the loss or altered speech and inability to eat/ swallow normally. Communication impairments may be due to aphasia,

dysarthria, or apraxia of speech [6]. Dysarthria and apraxia of speech represent the defects to the motor aspects of speech production [6], [7].

Clinical assessment of dysarthria

Changes in one's speech might be among one of the early signs of a neurological disorder [8]. Dysarthria - a motor speech disorder – is due to weakness, slowness, or incoordination of the musculature used to produce speech [6], [9]. Dysarthria is the most common occurring speech disorder in neurological conditions, with observed occurrence of 51% in patients with multiple sclerosis, 20-30% in individuals post-stroke, and 50-89% of PD patients [6]. Speech difficulties are observed at disease onset in approximately 30% of patients with Amyotrophic Lateral Sclerosis (ALS - aka bulbar onset ALS); while in 80-95% of ALS patients, dysarthria is developed as the disease progresses [10].

Auditory-perceptual assessment methods are considered gold standards for clinical differential diagnosis, assessing severity of symptoms and functional changes as well as making treatment plans and management decisions in dysarthria [11]. For example, a system developed by Darley, Aronson, and Brown established 38 perceptual dimensions defining auditory-perceptual changes in respiration, voice, articulation, prosody, and other aspects of speech to characterize the dysarthria type and localize the lesion site [11]. Auditory-perceptual methods have great advantage such as convenience, economy, and robustness, however, they have been also criticized for being susceptible to various sources of error and biases[12]. This method is heavily subjective and factors such as listener's shifting internal standards, past experiences and area of expertise, and type of rating scale used may become confounding factors [13].

Assessment of orofacial impairments

Orofacial muscles are often impaired in neurological diseases. For example, patients with PD can suffer from hypomimia, also known as masked face, in which they have reduced range and speed of facial muscle movements. In order to diagnose this condition, neurologists use the Movement Disorder Society – Unified Parkinson's Disease Rating Scale (MDS-UPDRS) to evaluate the presence and the severity of impairments. Section III of the assessment includes items that specifically evaluate the patients' facial expressions [14]. ALS patients have various orofacial

and speech difficulties that are observed in 80-95% of patients as the disease progresses with reduced speaking rate and speech intelligibility being identified as the worst aspects of the disease [10]. The ALS Functional Rating Scale – Revised (ALSFRS-R) is a 12-item questionnaire that allows patients to rate their current motor functions including speech, swallowing, and saliva management. It is a highly reliable and easily administrable system that have been used by clinicians to keep track of disease progression [15]. In stroke patients, the orofacial impairments depend on lesion's size and affected cerebral structures. Orofacial impairments may occur due to impairments of the cranial nerves, central nervous system pathways, or motoneuron pool. Stroke patients have demonstrated reduced tongue and lip forces. Facial asymmetry is also common in this clinical group [16].

The cranial nerve examination aims at providing information regarding the focal point of the pathology in the brain or brainstem [17]. It includes examining the functionality of the nerves that innervate muscles of the head and neck. The cranial nerves can be separated into four major groups: the sensory nerves, nerves responsible of eye movements, nerves of oral cavity and superior pharynx, and those responsible of neck movements. The neurological examination of these nerves includes palpation, check for symmetry and wasting, and testing for weaknesses or paralysis of muscles. The examination includes assessment of structures at rest, during nonspeech activities for example during sustained postures and movements as well as during speech tasks, and in response to reflex testing [18]. However, studies have shown that the assessments performed by the experts are subjective and heavily dependent on the clinician's expertise. Further, there are no agreed upon standards for detection of various disorders and monitoring of disease progression among experts in the field [15], [19], [20].

Using technology to support clinical assessment

With the advent of technology and increased availability of sensor-based solution to the consumer marker together with an attempt to overcome the limitations of clinical assessments, instrumentation-based protocols have been explored for the assessment of speech and orofacial impairments [21]–[23]. Some of these instrumented systems include those based on acoustic analyses (e.g., Phonatory aerodynamic system (PAS), nasometer) or kinematic/ movement-based methods (e.g., 3D optical motion capture system, and 3D electromagnetic articulography (EMA)).

Kinematic methods are of particular interest because they allow precise measurements of the movement associated with speech and orofacial functions.

The 3D optical motion capture system uses infrared light to track radio reflective markers that are placed on specific facial landmarks to track the facial movement during speech and non-speech tasks. The facial kinematics data such as range of motion and speed of each articulator provides valuable insight on the performance of the lips, jaw and other facial muscles [22]. This technology however requires a direct line-of-sight between the marker and the camera and cannot be used for the assessment of tongue. As a result, electromagnetic devices, technology based on electromagnetic field that does not require direct line-of-sight, have been used to track and assess the movement of the tongue as well as the lips, jaw and face during speech and orofacial assessments [24]. Studies using EMA have shown promising results and higher sensitivity to the early subtle changes/symptoms in patients relative to healthy adults compared to both clinicians and patients' self-reports [21], [25]–[28]. However, such systems have their own challenges that hinder their adoption into clinical settings including: their complexity and need of further specialized training for clinicians to be able to analyze and understand the data, the high cost associated with the instrumentation, and the long setup time of the sensor-based systems that result in patient fatigue and prolonged assessment time [15].

Solution: 3D cameras

The rapid development of 3D sensing technology in the past decade has resulted in many consumer applications. The Microsoft KinectTM was among the first 3D sensing consumer products released that enabled new gaming experience. This product used a depth sensor developed by Primesense that was based on structured light technology. Many other 3D sensing devices shortly followed, some examples were Intel RealSense 3D camera range, Intel® Euclid™ (Intel Corp., Santa Clara, CA, USA), Creative Senz3D (Creative Technology Ltd., Singapore), Asus Xtion Pro Live (AsusTek Computer Inc., Beitou District, Taipei, Taiwan), OptiTrack MoCap (NaturalPoint, Inc., USA), Vicon Infineon 3D Image Sensor (Vicon Motion Systems Ltd., Oxford, UK), and Kinect V2 (Microsoft, Redmond, WA, USA) [29]. As a result, various applications started integrating these technologies, such as gesture and action recognition [30], [31], facial expression analysis [32], and marker-less motion capture [33] just to name a few.

Intel RealSense 3D camera was one of the first low-cost, compact with high image quality that was widely adopted in the research community. Intel RealSense 3D camera consists of a depth camera (a projector and a sensor for triangulation) and an RGB camera that is full HD with 1920×1080 pixels resolution. The projector emits 860 nm near-infrared range light, and the IR sensor has a 640×480 pixels resolution with frame rate of 600 fps (1200 fps in half resolution). It is capable of providing a VGA-size depth map at 60 frames per second with 0.125 mm depth resolution, i.e., submillimeter accuracy [34].

There are numerous studies that have explored the use of such 3D cameras for clinical applications. In rehabilitation, such technologies have been considered for enabling patient feedback and correction while performing exercise regimens. For example, Chhor et al. [35] developed a computer game that used Intel RealSense 3D camera to aid upper body rehabilitation by tracking hand gestures. Other examples of applications using Intel RealSense 3D camera by tracking facial expression, movement of joints and skeletal tracking [29], [36], [37]. The depth sensing technology has also been used in facial analysis for tasks such as emotion recognition [38] and providing outcome measures for the assessment of neurological disorders in various clinical populations [39]–[43]. It is the latter application that has a great potential for integration into everyday clinical practice – for in clinic and remote assessments – as well as into clinical trials of pharmaceutical and behavioural interventions.

Facial Landmark Detection

The development of modern artificial intelligence (AI) and deep learning techniques – specifically, automated video analysis algorithms – afford substantial advances using video cameras for clinical assessment and monitoring. Facial landmark detection and tracking algorithms attempt to automatically detect the location of key facial landmarks on facial images or videos and are the building block of many applications including facial recognition, animation, tracking, expression analysis, and 3D facial modeling. Facial landmark detection is a challenging task for reasons such as diverse facial appearances across subjects, expression and pose changes of a given subject, partial occlusion and environmental factors (e.g., illumination). Therefore, earlier algorithms aimed at detecting landmarks in controlled conditions, for example under a fixed laboratory setting, with fixed head pose and minimal or highly specific facial expressions. In recent

years, more research has been focused on in-the-wild conditions, where algorithms aim at having a robust performance for arbitrary facial expressions, head poses, environmental conditions, etc. [44], [45].

Facial landmark detection algorithms are generally supervised; they use a large set of manually labelled training facial images, from which the AI model learns to locate the same landmarks on future test face images. These models first use a face detector to obtain a rectangular bounding box of the face within a given image. Depending on the model, different number of facial landmarks are outputted (e.g., a 17-point model, 29-point model, or 68-point model) marking the locations of the eyes, nose, mouth etc. on the face. The output is in a form of vector $x = (x_1, \ldots, x_N, y_1, \ldots, y_N)^T$ where (x_i, y_i), which denotes the pixel values of the ith landmark in the image, refer to Figure 1-1.

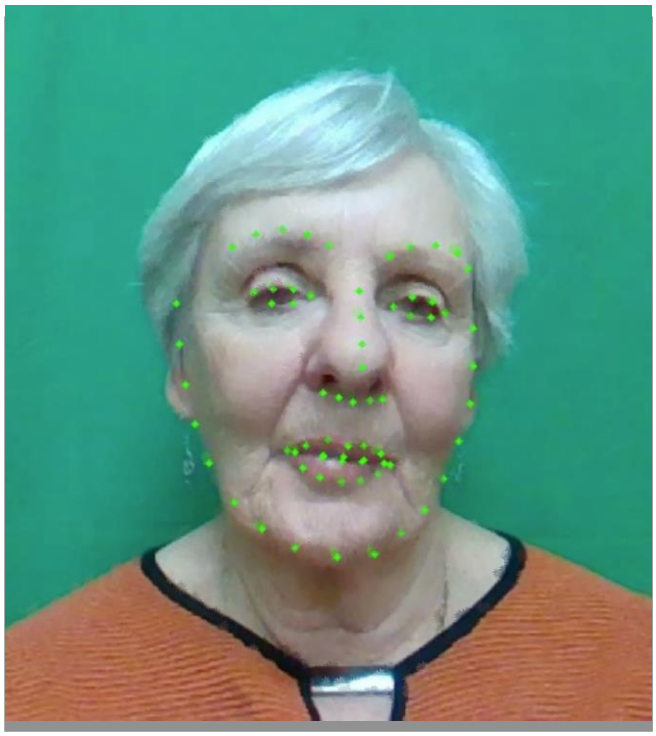

Figure 1-1 Output of facial landmark detection algorithm, the pixel values of x and y coordinates are superimposed on the input face image

Facial landmark detection models prior to deep learning are categorized into two primary categories, based on either parametric or nonparametric methods. The parametric methods are built based on data belonging to a particular distribution such as a Gaussian mixture model, while

nonparametric models do not assume a distribution. The parametric models have a fixed set of parameters that are tuned based on the input data. Gaussian mixture models and the Active Appearance Models are the most common methods used in this category[46]–[50]. In contrast, the number of parameters in the nonparametric models may vary based on the number and complexity of the training data; these models may still have implicit assumptions or shape constraints built in because of the specifics of the training data. Graphical models such as tree structures and Markov random field methods, in which the location of each facial landmark is optimally solved using dynamic programming, are examples of nonparametric methods [51].

In recent years, deep networks, e.g., convolutional neural network (CNN), have shown superior performance for facial landmark detection. These methods learn the facial shape and appearance variations and the nonlinear mapping from facial images to landmark coordinates. Examples include deep hierarchical face parsing [52] and coarse-to-fine autoencoder networks [53]. In general, deep neural net-based algorithms are divided into two main categories of direct regression methods that directly predict the coordinates of each landmark (e.g. [54]–[58]) and heatmap regression-based methods, where a 2D heatmap is generated for location of each landmark (e.g. [59], [60]) [61]. Some of the common architectures used as building blocks for either of the above-mentioned approaches are ResNet [58], MobileNet [57], ShuffleNet [56], Hourglass [59], HRNet [62], and CU-Net [60]. The most commonly used architecture is hourglass architecture, initially introduced for pose estimation [63], which is an encoder-decoder network block that can be sequentially stacked [59].

In facial video analysis, recurrent neural networks (RNN) [64] and long short-term memory (LSTM) [65] architectures have been used to minimize frame-to-frame landmark jitter. Based on a review conducted on the state-of-the-art facial landmark detection algorithms developed in recent years [61], some of the best performing models on various datasets include SubpixelHeatmap [66], ADNet [67], and PropagationNet [68]. In SubpixelHeatmap, the authors use local soft-argmax computation to reduce the quantization error of the generated heatmap [66]. The authors for ADNet use the fact that the error in the tangent direction (relative to face boundary) in general has higher value than the error in the normal direction, thus, design the architecture such that it assigns higher weights to normal error [67]. In PropagationNet, the authors note the importance of facial boundary for accurate detection of the facial landmarks [68]. In this approach, convolutional

blocks transform landmark heatmaps into boundary heatmaps after each Hourglass module and use it as an intermediate attention mask to improve final accuracy predictions [68].

There are several publicly available datasets for training facial landmark detection models [69]–[72]. These datasets can be categorized into those captured in controlled conditions (e.g., in a lab setting with fixed head pose and facial expression) [71], [73], or those captured in uncontrolled environments, i.e., in-the-wild [70]. The latter are often collection of images from websites such as Facebook or Google images. The ground truth facial landmarks are obtained manually by employing workers or crowdsourcing through services such as the Amazon Mechanical Turk. Each image is generally labelled by several workers and the averaged labelled values are taken as final ground truth of landmark positions. The accuracy of the models is measured by computing the distance between the landmark positions predicted by a landmark detection model and the ground truth labels. The computed distance is commonly normalized by the interocular distance and is referred to as the normalized mean error (NME) [61]. Another metric used to report model performance is the failure rate (FR), which denotes the number of images with NME above a given (e.g. 10%) threshold [61].

Facial landmark detection and tracking is an important first step (i.e., a preprocessing step) for many application areas such as face recognition, expression and facial movement analysis, as well as pain detection [74]–[78]. Exploring various algorithms and deep learning architectures and methodologies to improve the tracking accuracy remains an active area of research. Currently, the focus of the field is to balance accuracy and complexity of the models being developed as well as leveraging semi-supervised and unsupervised learning methods in order to take advantage of the abundant unlabeled content online to more effectively train the models [79].

The Face Alignment Network

The Face Alignment Network (FAN) [80] is a deep neural network with residual blocks trained on a large pool of real and synthetically expanded images with ground truth facial landmark locations. The FAN architecture is composed of four sequential components, each with an hourglass (HG) architecture [63], in which the bottleneck blocks were replaced with hierarchical, parallel, and multiscale blocks [81]. The outputs are 68 heatmaps of 64×64 dimensions, each of

which estimates the position of a facial landmark. The 2D-FAN was trained using 230,000 images in the training set and was evaluated on 7,200 images in the test set [80].

In previous orofacial analysis studies, FAN was used to extract facial landmarks in clinical populations [82]–[85]. Aligned with findings from Asgarian et al. [86], these studies found the presence of biases resulting in reduced landmark localization accuracy in clinical populations compared to healthy age-matched controls [82]–[85]. As a result, a series of studies addressed the need to reduce such biases via fine-tuning the FAN on data from specific patient groups. Bandini et al. released a publicly available dataset, the Toronto NeuroFace Dataset, from which more than 3,300 manually annotated images of faces from individuals with ALS, post-stroke, and aged-matched controls were used to fine-tune FAN [73]. In follow up studies, Guarin et al. showed that fine-tuning FAN using the clinical data improved the overall model's performance [84] as well as improving the performance of subsequent classifiers using detected landmarks in differentiating clinical groups from healthy controls [87]. Guarin et al. further quantified the amount of clinical data needed to fine-tune and improve the performance of FAN [85]. It was found that fine-tuning the model with as little as 8 annotated images from a single patient resulted in better performance on other individuals from the same clinical group. The best performance was achieved by fine tuning the model with 320 images from 40 patients and using more images from additional patients did not improve the performance further. Overall, the findings showed that fine-tuning FAN using a relatively small sample of clinically represented data improved the performance of the model as well as the subsequent clinical facial analysis task.

Validation Framework

There have been frameworks developed specifically for the digital health technology development and evaluation [88]. The V3 framework describes three phases for the evaluation of digital biomarkers, namely, verification, analytical validation, and clinical validation [89]. According to this framework, verification refers to validating the hardware and sensors involved in recording the digital measurement. Analytical validation refers to the process of ensuring that the digital biomarker measurements are accurately measuring the intended phenomena. Clinical validation is concerned with evaluating whether the digital biomarker provides meaningful clinical information [89].

Robin et al. specifically examined this framework and provided further recommendations of how to use this framework for speech related biomarkers and speech-based measures [90]. Using this framework, the verification step involved comparing various technologies and establishing the adequacy of the final technical instrumentation at collecting parameters that can be acceptable digital biomarkers [90]. The verification step of any technology is usually performed by the manufacturer via bench top testing [89]. Analytical step varies depending on the relevant features or methods used to compute the biomarker. This method can entail comparing the new digital biomarker to the existing technical gold standards, obtaining concurrent validation through comparison with other sensors, or through other methods such as cross-validation and independent replication of results. At this phase it is also necessary to analyze and eliminate potential biases in the data as a result of the confounding factors.

Clinical validation is the culmination of technology evaluation. At this stage, the technology outputs are compared with clinical gold standards that have been established to detect disease and track its progression. In the assessment of various neurological disorders, some diseases have a clear gold standard while for many others do not. Therefore, in clinical validation, the choice of reference measure limits the clinical validation as the digital biomarker can only be as good as the measure used to validate it. Clinical validation can be especially challenging, when the used reference measure has limited validity, or lower reliability, resulting in difficulties achieving consistency [90]. Using this framework, engineers, scientists, and clinicians must work together to bring a technology from a design and prototype phase into a clinically utilizable tool [90].

Current Technology: Verification

The proposed system used in this thesis consists of Intel RealSense 3D Camera to collect data from patients performing standard orofacial tasks as well as use of the FAN model to automatically extract facial landmarks that are then used to further extract facial kinematics. The performance of both components (the Intel RealSense 3D Camera and FAN model) has been previously studied and verified by their developers as well as the research community. Keselman et al. [91] performed a comprehensive study on the Intel RealSense 3D camera and verified the performance of the system under different test conditions. Performance of FAN was tested on

some of available open-source datasets and its performance was reported relative to the other facial landmark detection methods, as well as relative to manually annotated facial landmarks [80]. Together the system made from combining these two technologies has the potential to provide clinicians with an objective way of analyzing facial kinematics. However, this system needs to analytically and clinically validated before its adoption as a clinical tool, which are main goals/objectives of this thesis.

Current Technology: Analytical Validation

The analytical validation process evaluates the generated physiological and/or behavioural metrics in experiments involving human subjects [89]. The Intel RealSense 3D Camera has been previously analytically validated in a study where, Ouni et al. [92] compared Intel RealSense 3D against the sensor-based Electromagnetic Articulography (EMA) and found the tracking error of the lower lip to be low and the system suitable for facial applications that requires facial motion tracking. In another study, Bandini et al. [93] compared the Intel RealSense 3D Camera along with Intraface facial landmark detection and tracking algorithm against the marker-based optical tracking system and found the lips Root Mean Square Error (RMSE) to be between 1-3 mm range. Harkel et al. [94] have conducted a reliability study comparing Intel RealSense 3D Camera against a highly costly and accurate 3D facial tracking system and found high agreement between the two methods in patients with unilateral facial palsy. The accurate tracking of the facial landmarks is a prerequisite to performing clinically meaningful facial analysis. The development of new and more accurate architectures remains an active area of research and their adoption in developing objective clinical tools is on the rise [95]. However, according to the V3 framework, new analytical validations are needed when there are changes/upgrades to either the software and/or hardware [89]. Thus, as there were no analytical validation studies done previously that uses the Intel RealSense 3D Camera together with the FAN to analyze facial kinematics, the first study of this thesis aims at evaluating this technology.

Current Technology: Clinical Validation

Clinical validation process evaluates whether the proposed technology provides any valuable clinical insight in a specific context [89]. Clinically, the primary goals of an assessment are to detect clinical features that point to the presence of an abnormality that can lead to disease

diagnosis and/ or allows tracking the disease progression over time. Clinical gold standards carry these two important missions – detection/ diagnosis vs. disease tracking [96], [97]. The proposed technology has been previously validated in automatic assessment of facial video recordings during speech and non-speech tasks to distinguishing healthy vs. post-stroke [39], healthy vs. Parkinson's disease [41], and healthy vs ALS individuals [23]. Several facial landmark detection and tracking algorithms have also been clinically validated in other applications such as: automatic pain detection from facial images or videos of participants [98], [99], automatic detection of patients with neurological disorders (e.g. ALS, stroke, PD, and AD) from healthy participants using facial kinematics [39], [42], [100]–[103], and automatic detection of psychiatric conditions (e.g. depression, and schizophrenia) [77], [104], [105]. In all these studies, the main goal was to extract clinically interpretable objective facial features from the landmarks such as facial dynamics and kinematic measures and then to use these measures to detect the presence of disease via classification methods.

Another important goal of these measures is to track disease progression over time. For that, correlations with clinical gold standards must be established. The studies predicting the severity of symptoms have been limited, but important in the effort to clinically validate novel assessment technologies. One study showed that kinematics of facial expressions was able to predict the depression severity estimated via the Beck Depression Inventory-II (BDI-II) [105].

The goal of this thesis is to further clinically validate the use of specific set of deep learning algorithm (FAN and fine-tuned FAN) together with the Intel RealSense 3D camera for use in the clinical setting for detecting severity of orofacial impairments within clinical groups.

Summary: Motivation and Gap

Detection of most neurodegenerative diseases are highly dependent to the clinicians' experience and are not well standardized [15]. This may lead to delayed diagnosis, and therefore, delayed interventions with reduced possibilities to slow down the rapid reduction of quality of life for the patients. Recent advances in computational power and deep learning algorithms have provided an opportunity to develop potential solutions for automated assessment and early detection of neurological disorders in clinic and remotely, in home settings.

Previous studies have established the Intel RealSense 3D camera's technical verification and shown its high accuracy and performance on bench top experiments [34], [91]. Some studies have also shown analytical and clinical validation of the Intel RealSense 3D camera [29], [43], [73], [94]. Other studies have used facial landmark algorithms to perform clinically relevant facial analysis [41], [42], [73], [106]. However, the combination of the Intel RealSense 3D camera with FAN to perform objective orofacial assessments in clinical groups with facial impairments have not been fully evaluated in a context of V3 framework before. This research thesis focuses on the analytical and clinical validation of the proposed technology (the Intel RealSense 3D camera and FAN) with the aim of advancing its translation into clinical settings.

Research Question and Objectives

The ultimate objective of this research is to implement an affordable, clinically valid, easy-to-use video-based application to support the detection and assessment of orofacial deficits in neurological diseases in order to support clinicians with the disease diagnosis and prognosis. The objective of this thesis is to analytically and clinically validate the use of an accessible 3D camera and a machine learning-based facial landmark tracking method for extracting kinematic facial features that are representative of the perceptual clinical scores provided by trained clinicians to individuals with orofacial impairments due to neurological disorders (e.g., post-stroke and ALS). In study 1, analytical validation was determined by comparing the kinematic signals extracted from the proposed system (the Intel RealSense 3D camera and FAN) against the gold standard technology – a sensor-based system EMA. In study 2, clinical validity was determined using correlations between the objective kinematic features and perceptual clinical assessments performed by an expert clinician. In the first study (Chapter 2) regarding analytical validation, we hypothesize that the newly proposed system would perform as well as the accepted gold standard method (EMA) with high correlation between the two methods. In the second study (Chapter 3) about clinical validation, we hypothesized that the kinematic features and corresponding perceptual clinical scores – symmetry, range of motion (ROM), and speed – would be significantly correlated with corresponding clinical scores and could be used to predict the severity of oromotor impairment.

Structure of This Document

This thesis is structured as follows:

Chapter 1 provides a general introduction to the research topic as well as the rationale for this study. It provides an overview of the related literature for the technologies used in this thesis for performing objective orofacial assessments of neurological disorders, including, 3D cameras, AI-based facial detection and tracking algorithms, and application of such algorithms in detection of orofacial impairments. It also lays out the problem, the motivation behind the research, the research framework and objectives, and contributions of this work.

Chapter 2 is the manuscript describing the analytical validation of video-based technology for assessing orofacial movements

Chapter 3 is the manuscript describing the clinical validation of these technologies in assessing orofacial impairments in clinical groups

Chapter 4 provides an in-depth discussion of the obtained results, study limitations as well as outlines potential future directions for this research.

This research has been submitted to:

(1) Chapter 2: Jafari, D. Bandini, A. Guarin, D. Taati, B. Yunusova, Y. Validation of Markerless Video Tracking of Lip Movements for the Assessment of Orofacial Functions. [Submitted to 15th IEEE International Conference on Automatic Face and Gesture Recognition]

(2) Chapter 3: Jafari, D. Guarin, D. Simmatis, L. Bouvier, L. Taati, B. Yunusova, Y. 3D Video Tracking Technology in the Assessment of Oro-facial Impairments in Neurological Disease: Clinical validation. University of Toronto, Toronto, 2022 [Submitted to the Journal of Speech, Language, and Hearing Research]

Chapter 2 – Analytical Validation

2.0 Abstract

Facial movements provide valuable information for the assessment of orofacial functions such as speech, swallowing, and facial expressions. Due to the high cost and cumbersome setup procedures, sensor-based technologies have not been adopted in the clinical settings. With the recent advancements in markerless facial tracking algorithms as well as developments of 3D cameras, it has become possible to develop viable clinical applications for the diagnosis and treatment of motor speech disorders. In this study, we assessed the performance of a video-based system composed of a deep neural net facial tracking algorithm coupled with Intel RealSense camera to track the lip movements. The validation was performed on three healthy control participants performing a set of speech and non-speech tasks similar to a routine orofacial assessment conducted in clinical settings. Preliminary results show moderate to strong correlation between this method and the gold standard sensor based electromagnetic articulography. Further algorithm training and tests are needed in order to improve the performance of such system in order to develop an accurate clinical tool for the assessment of orofacial movements.

2.1 Introduction

Accurate assessment of orofacial movements is important for the diagnosis and treatment of various neurological disorders. Changes in the movement of the lips and the jaw have been associated with early signs of oromotor decline, disease progression, and speech intelligibility decline in neurological diseases such as amyotrophic lateral sclerosis (ALS), Parkinson's disease (PD), and stroke [107], [108]. Sensor-based technologies such as electromagnetic articulography (EMA) and optical tracking systems are well established methods for the quantitative assessment of orofacial movements. However, such technologies are not clinically feasible due to their high cost, system complexity, and long and tedious setup procedures. As a result, the quantitative analysis of orofacial movement has been limited to research purposes and not integrated into clinical practice in neurology or speech-language pathology [23], [40], [92], [109]. There is a need for an inexpensive, easy-to-use, and clinically feasible orofacial movement tracking system that is able to accurately and quickly quantify patient performance in a clinical setting. Recent advances in computational power and deep learning algorithms coupled with the development of 3D depth cameras have provided an opportunity to develop affordable, markerless, automated, and clinically objective solutions for the assessment of neurological disorders affecting orofacial functions [41], [93], [110], [111].

The evaluation of 3D cameras as potential tools for facial tracking has begun by the research community. Ouni and Dahmani assessed the performance and precision of two 3D markerless systems (Primesense and Intel RealSense cameras) with respect to an optoelectronic marker-based system in sentence and syllable productions. They used the commercial software Faceshift to track the virtual markers manually placed on the 3D mesh of the facial model in a post-processing step. The outcome of the study proved overall good performance and low RMSE for both markerless systems, with Intel RealSense 3D depth camera being slightly more accurate than the Primesense (RMSE < 2 mm) [92]. Bandini et al. evaluated the accuracy of Primesense depth sensor alongside InteraFace facial landmark tracking software package [112] for 3D tracking of lip movements in various speech tasks with RMSE range of 1-4 mm [93]. In a later study, Bandini et al. were able to discriminate people with PD from healthy control participants using lower lip movement during syllable repetitions under the same video-based settings and confirmed that patients have reduced articulatory kinematics [41]. Bandini et al. also validated the video-based tracking of jaw movements with Intel RealSense SR300 camera and the RealSense SDK R3 face tracking algorithm against the EMA during speech with the similar accuracy (average RMSE ~ 2 mm) [40]. Other preliminary work has focused on and shown high promise for utilizing open-source video-based facial landmark tracking algorithms in clinical assessments of individuals with orofacial deficits, demonstrating high accuracy (up to 87%) in detecting orofacial impairments [110], [111]. As such, the role of computer vision research to study human face for clinical applications has thrived over the past few years, with many applications in neurological diseases [41], [98], [110], [111]. Despite a significant promise to revolutionize neurological assessment, further research is needed to validate the use of deep learning facial landmark tracking models in assessing orofacial movements. Markerless methods use various computer vision algorithms and machine learning models to retrieve and track facial landmarks in video data. As a result, the quality of the tracking is highly dependent on the tracking model used [92]. The goal of this study was to evaluate the feasibility of state-of-the-art markerless video-based system, as compared to the gold standard EMA for tracking lip movements. The markerless system consisted of a pre-trained convolutional neural network (CNN) model for facial landmark tracking as well as a 3D depth camera registered with the video camera in order to translate 2D facial landmarks tracked by the CNN model into 3D locations.

2.2 Method

2.2.1 Experimental Setup and Procedure

Three healthy young female volunteers were recruited for the experiment. The demographic details of participants is provided on Table A-1 in Appendix A1. The participants were recorded via 3D (video and depth) camera as well as EMA system while performed a series of speech and non-speech tasks typically used by clinicians in a routine assessment of orofacial and speech function. The tasks were expanded to include conditions (e.g., fast, slow) which are known to enhance performance variability within a speaker. The tasks and conditions were:

Sentence repetition: The sentence "Buy Bobby a Puppy" was repeated five times at normal and slow speaking rates, both with normal loudness.

Max opening: The mouth was opened to the maximum comfortable range and then closed to the rest position, five times. Four variations of the task were performed, namely, (1) Hold: where participants held the maximum opening for five seconds, (2) Normal: maximum opening followed by closing at a normal speed, (3) Fast: the task was performed as fast as possible, and (4) Asymmetric: opened the mouth asymmetrically to one side (left or right).

Max spread: The lips were spread out to the sides without opening, starting from the rest position without a pause (Normal), or with five seconds pause (Hold) at the maximum spread, repeated five times.

The recordings were performed in a quiet room, with a single constant and uniform light source placed behind the camera. Participants were seated during the experiment looking straight at the camera. Head movements were monitored and large head movements during experimental recordings were avoided. The EMA sensors were placed using clear surgical tape on the participants' face prior to data acquisition, and the camera was adjusted to be at each participant's eye level. The Wave electromagnetic generator was located on the right side of participants' head, while the 3D camera was placed at a distance of 0.3-0.4 m away from the tip of the nose.

2.2.2 Data Acquisition

1) The Wave speech research system (NDI – Waterloo, Ontario, Canada) was used to track lip movements. The Wave system consists of an electromagnetic field generator and sensors that report 5 degrees-of-freedom position information within the set boundaries of the generated EM field with high tracking accuracy (average error = 0.8 mm), thus, making it the "gold standard" for validation of video-based systems [109]. Eight sensors were attached to the participants' faces in the following positions:

N1: placed at the intersection of the frontal and nasal bone

N2: placed at the tip of the nose.

LU (LB): placed at the upper and the vermilion of the upper and lower lips, respectively.

LR (LL): placed at the right and left corners of the mouth, respectively.

JR (JL): placed on the inferior of the jaw about one quarter of the distance from the gnathion and the right (left) gonion.

The 3D position of the sensors over time (trajectories) were collected at the sampling frequency of 100 Hz. Data were recorded and stored using the Wavefront software v.1.1.

2) The video-based markerless system consists of an Intel RealSense SR300 (Intel Corp., Santa Clara, CA, USA) camera that acquires registered color and depth video recordings, as well as the FAN model [80]. FAN is an open-source CNN model used to automatically detect the face, localize 68 facial landmarks, and track them throughout the duration of the videos.

The videos were collected at the resolution of 640×480 pixels at 30 frames per second (fps) rate with color (RGB) and depth recording occurring simultaneously. Videos were recorded using custom C++ code.

2.2.3 Data Analysis

1) Pre-processing: A custom Python script was developed to process and analyze data obtained from each of the two data recording methods used. Because both EMA sensor recordings and the video recordings randomly drop a few data points resulting in inconsistencies within the sampling rate, both data streams were resampled at a uniform rate of 100 Hz using linear interpolation. A low-pass Butterworth filter (8-poles, with cut-off frequency of 15 Hz) was used on the trajectories of the points of interests (landmarks and sensor trajectories) to eliminate jitter.

2) Kinematic measures: As interpretable decisions are crucial in clinical applications, the measures of range of motion and average speed, aligned with typical clinical judgements, were selected for the comparisons.

The range of motion (ROM) was estimated vertically and horizontally. The vertical range of motion (VRM) was calculated as the Euclidean distance (in mm) between LU and LB (Wave) and the distance between landmarks 52 and 58 Figure 2-1. The horizontal range of motion (HRM) was calculated as the Euclidean distance between lip corners - RC and LC (Wave) and the distance between landmarks 49 and 55 (refer to Figure 2-1). The velocity signals (in mm/s) were computed as the first derivative of the calculated displacements with respect to time.

3) Validation: The phase correlation in Fourier space [113] was used to align the signals obtained from each method for all the recordings. The zero-lag cross correlation (CC0) was computed to measure the agreement between the two signals obtained from each of the methods for both displacement and velocity features extracted. These measures were calculated for each task and then averaged across the three participants. The tasks were then parsed into individual repetitions and the maximum value of ROM and average speed were calculated for each repetition of the two methods used. The Spearman's correlation (SC) coefficient was then calculated for each task across all the repetitions for each participant and across the three participants.

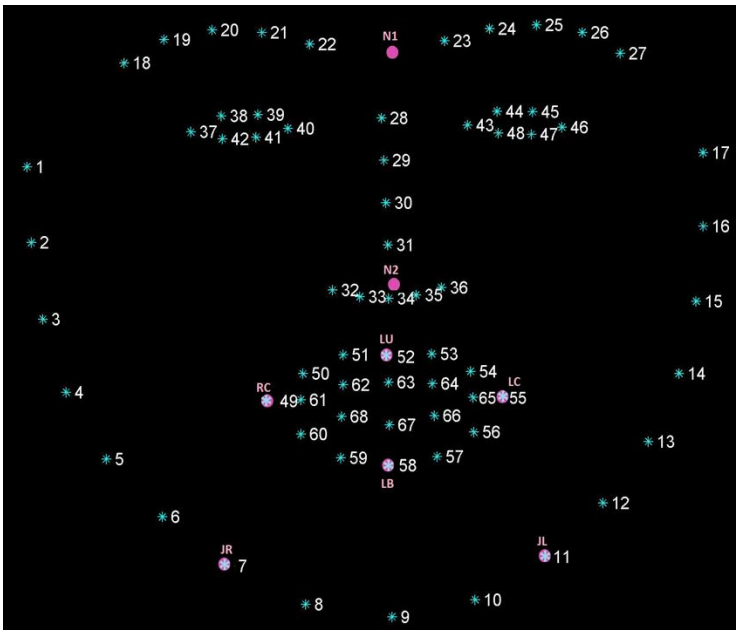

Figure 2-1 Position of EM sensors (purple dots) and facial landmarks (blue asterisks)

2.3 Results

Figure 2-2 shows an example of VRM of the mouth during the max opening and sentence repetition tasks followed by HRM during the max spread task. Table 2-1 summarizes the CC0 for the entire signals averaged across the participants. Table 2-2 summarizes the SC coefficients for the maximum value of ROM and average speed of the repetitions across each task per participant as well as across all three participants (Total). Figure 2-3 shows the max ROM and the average speed values of the repetitions for the max opening task. Figure 2-4 shows the max ROM values for the max spread task.

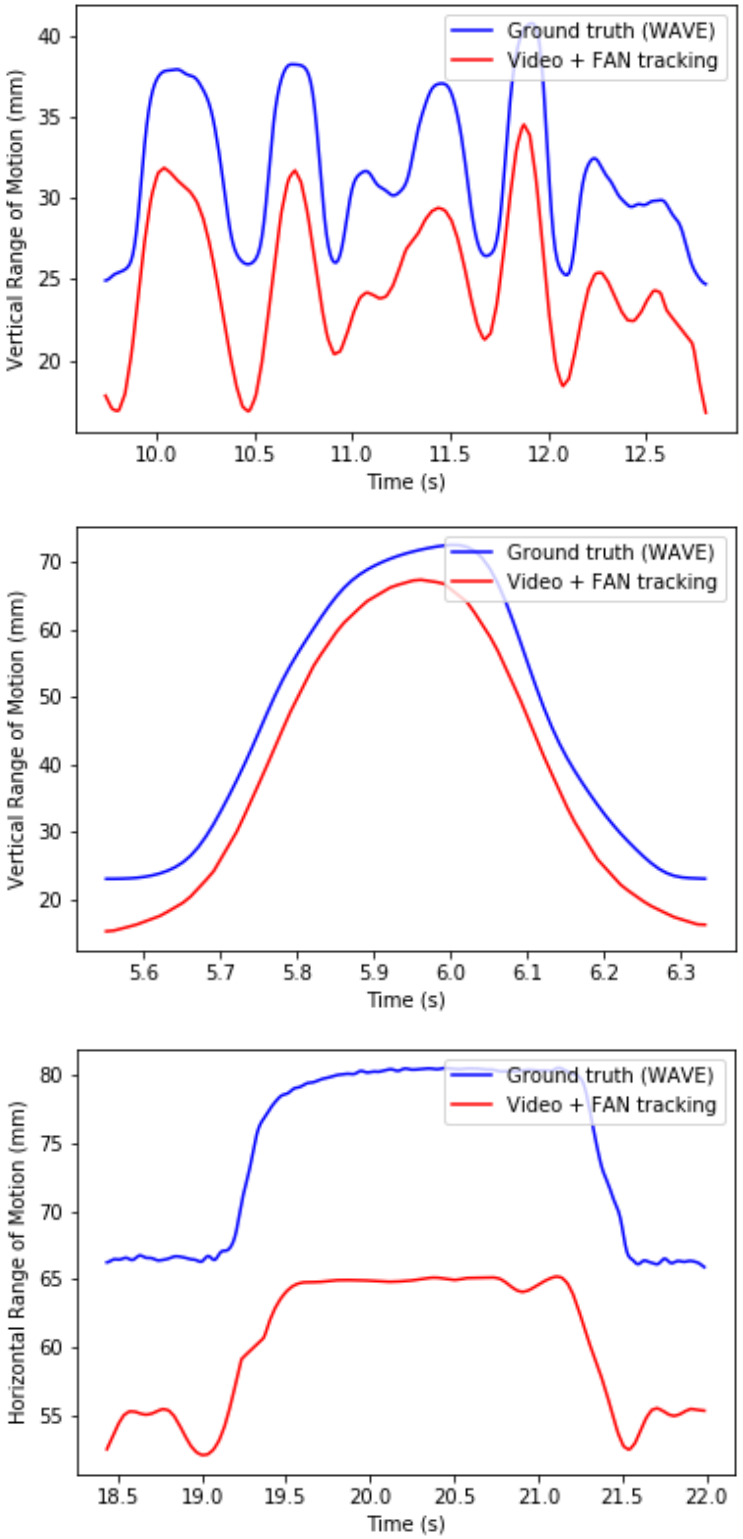

Figure 2-2 VRM signals during the speech repetition (top) and the maximum open (middle) tasks and HRM for the maximum spread task (bottom) at the normal condition.

Table 2-1. Measures computed to assess the accuracy of the video-based method via zero-lang Cross-Correlations (CC0) of the entire rom or velocity signal

Task	ROM Measure	Mode	CC0 of ROM	CC0 of Velocity
Open	***VRM***	***Hold***	0.99 ± 0.01	0.83 ± 0.19
		Normal	0.91 ± 0.13	0.79 ± 0.21
		Fast	0.99 ± 0.01	0.94 ± 0.05
		Asymmetric	0.91 ± 0.08	0.71 ± 0.21
Spread	***HRM***	***Normal***	0.90 ± 0.04	0.29 ± 0.25
		Hold	0.90 ± 0.08	0.22 ± 0.26
Sentence	***VRM***	***Normal***	0.81 ± 0.07	0.64 ± 0.12
		Slow	0.85 ± 0.02	0.55 ± 0.16

Table 2-2 Spearman Correlations (SC) for rom and average speed computed per participant and total, per task and across conditions within each task

	Task	ROM Measure	SC of Max ROM	SC of Average Speed
Participant 1	***Sentence repetition***	***VRM***	0.92	0.55
	Max Opening	***VRM***	0.75	0.92
	Max spread	***HRM***	0.63	0.32
Participant 2	***Sentence repetition***	***VRM***	0.67	0.68
	Max Opening	***VRM***	0.31	0.88
	Max spread	***HRM***	0.53	0.12
Participant 3	***Sentence repetition***	***VRM***	0.37	0.77
	Max Opening	***VRM***	0.64	0.81
	Max spread	***HRM***	0.04	0.53
Total	***Sentence repetition***	***VRM***	0.57	0.54
	Max Opening	***VRM***	0.91	0.86
	Max spread	***HRM***	0.74	0.57

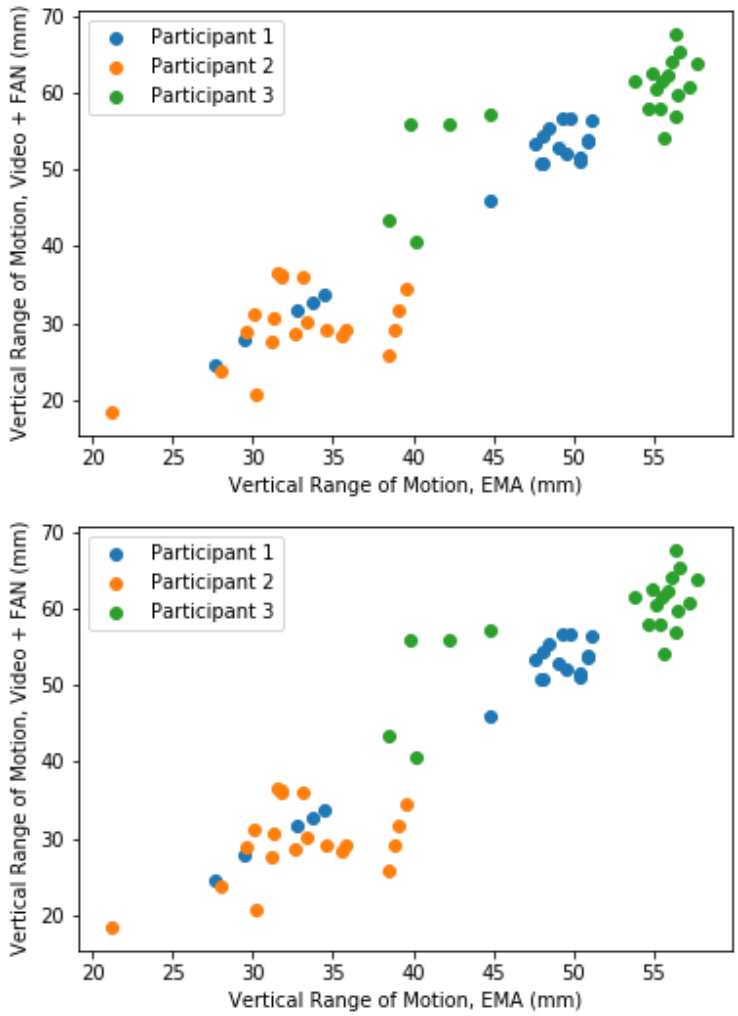

Figure 2-3 Max VRM (top) and Average Speed (bottom) measures per repetition for the three participants during the max opening task

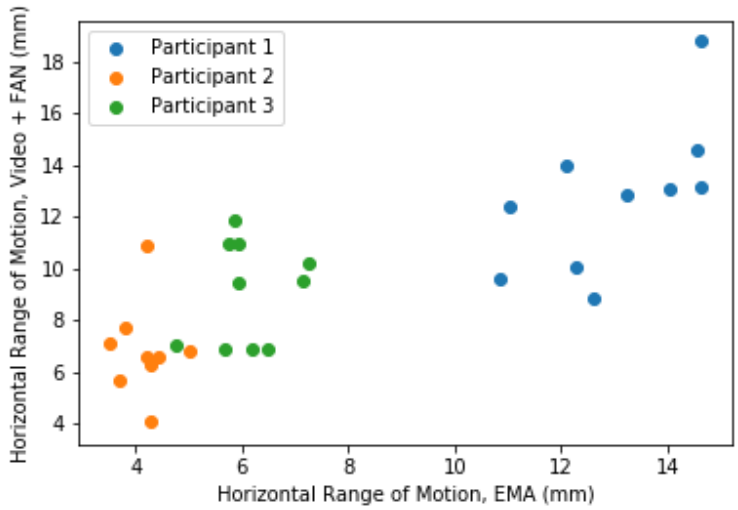

Figure 2-4 Max HRM (top) measures per repetition for the three participants during the max spread task

2.4 Discussion

This preliminary study showed high correlations between the EMA and video-based tracking systems with zero-lag cross-correlation values > 0.90 for non-speech tasks and >0.81 for the speech repetition task, when looking at the entire signals. However, the correlation values were smaller when extracting the maximum ROM and average speed values on signals parsed by repetition to obtain discrete, clinically relevant measures. This could potentially be due to the jitter in the facial landmarks obtained using FAN [80], particularly when calculating the correlation between velocity values. In future studies, additional signal processing or improved facial landmarks tracking will need to be considered to reduce frame-to-frame jitter. The correlation values were higher across participants as opposed to within each participant's repetitions of the task. This is due to the fact that the variability of kinematic values were low within each participants' repetitions compared to the trend across all the participant's data.

As illustrated in Figure 2-2, there was an expected bias in the signals obtained from the two methods. This bias was due to the fact that the two methods measured different distances: the ground truth signal (blue) was the distance between two sensor locations, while the FAN tracking signal (red) was the distance between two automatically detected facial landmarks (lip edges). The positions of these reference points were naturally not the same. In addition, some discrepancies between the signals are likely to be due to the sampling rate difference between the two methods, as the video were recorded at a lower sampling rate than the EMA data (30 fps compared to 100 Hz). Future work should aim to record data at the minimum sampling rate of 50 Hz, as recommended by Ouni and Dahmani [92], to enable fine modeling of lip movements.

Figure 2-3 shows that the video-based system was able to successfully differentiate between different conditions within individuals (i.e., as expected, the fast condition showed smaller ROM and greater average speed) for the maximum open task. Additionally, it was able to differentiate between speakers, as some participants naturally use larger or smaller movements than other participants. This differentiation is clinically relevant as clinicians are tasked to visually identify the movement ROM and speed. These perceptual judgements have been known to have reduced reliability, and the potential use of video-based methods to distinguish individual differences is highly promising.

Clear advantages of the video-based tracking have emerged. First, the acquisition volume for the EMA – determined by the Wave field generator placed near the participant's head – was smaller than the adjustable and larger volume of video-based tracking. In EMA, even small head movements away from the generator's field resulted in the loss of tracking by the sensors. We had to closely monitor the head position of the participants in the EMA field. However, such small movements were accounted for in the video tracking via automated face detection and the detection of facial landmarks so long as the face remained within view of the camera, thus, avoiding the complete loss of data. Furthermore, electromagnetic tracking was limited by the number of sensors placed on the face and highly reduced/limited movement information available for EMA, while the current state-of-the-art facial tracking algorithm tracked large number of facial features (e.g. 68 landmarks). Although we extracted relatively small number of highly straightforward features, the work on the most important feature is still ongoing [42].

In conclusion, this study has demonstrated that the video-based tracking using the FAN [80] has a great potential in tracking facial features relative to the gold standards. Further improvements to its ability to detect facial landmarks needs to establish across a large number of participants with and without neurological disease and across tasks. This ongoing work can help in accurate tracking of facial movements, thus, providing clinicians with guidance in the assessment of neurological disorders.

Chapter 3 – Clinical Validation Study

3.0 Abstract

This study sought to determine whether clinically interpretable kinematic features extracted automatically from 3 dimensional (3D) videos were correlated with corresponding perceptual clinical orofacial ratings in individuals with orofacial impairments due to neurological disorders. 45 participants (19 diagnosed with motor neuron diseases (MND), and 26 post-stroke) performed two non-speech tasks (mouth opening and lip spreading) and one speech task (repetition of a sentence "Buy Bobby a Puppy") while being video recorded in a standardized lab setting. The colour video recordings of participants were assessed by an expert clinician – a speech language pathologist - on the severity of three orofacial measures: symmetry, range of motion (ROM), and speed. Clinically interpretable 3D kinematic features, linked to symmetry, ROM and speed, were automatically extracted from video recordings, using a deep facial landmark detection and tracking algorithm, for each of the three tasks. Spearman correlations were used to identify features that were significantly correlated (p-value < .05) with their corresponding clinical scores. Clinically significant kinematic features were then used in the subsequent multivariate regression models to predict the overall orofacial impairment severity score. Several kinematic features extracted from 3D video recordings were associated with their corresponding perceptual clinical scores, indicating clinical validity of these automatically derived measures. Different patterns of significant features were observed between MND and post-stroke groups; these differences were aligned with clinical expectations in both cases. The results show that kinematic features extracted automatically from simple clinical tasks can capture characteristics used by clinicians during assessments. These findings support the clinical validity of video-based automatic extraction of kinematic features.

3.1 Introduction

Neurological disorders often affect orofacial and speech functions [6]. For example, people post-stroke or living with motor neuron diseases (MND) often experience orofacial motor impairments (i.e., inability to produce facial expressions or perform volitional orofacial movements), as well as dysarthria, a motor speech disorder [114], [115]. Accurate assessment of orofacial speech and non-speech movements becomes important for effective disease detection, tracking progression, and monitoring treatment outcomes [15].

Existing orofacial assessment methods rely on either subjective clinical evaluation performed by experts or on the use of complex and expensive sensor-based techniques, such as electromagnetic articulography (EMA), that are not suitable for integration into current clinical workflows. Emerging video-based methods may offer easy-to-use and cost-effective solutions to

support objective clinical assessment of orofacial motor impairments. The development of 3D video cameras and deep learning algorithms has enabled accurate automatic detection and tracking of pre-defined facial landmarks in facial images [80], which is the first essential step for many clinical facial analysis applications.

In order to be adopted into a clinical setting, novel facial tracking technologies must be evaluated within established frameworks for digital biomarker development, such as the FDA BEST (FDA, 2021) criteria and the V3 framework [89]. The V3 framework advocates for a 3-part sequential biomarker evaluation procedure: (1) verification of sensor-level outputs, typically conducted by manufacturers, followed by (2) analytical validation of a novel method with respect to an already-validated technical gold standards, followed by (3) clinical validation to assess utility in a specific clinical population and relative to existing clinical standards/ practices. The technical verification of the facial tracking technologies has been previously well established [29], [34], [91]. The analytical validation of markerless methods has demonstrated promising results as well. Ouni and Dahmani found that facial tracking methods using two different 3D cameras, namely, Intel RealSense (Intel Corporation, Santa Clara, California, USA) and PrimeSense Carmine (PrimeSense, Tel Aviv, Israel) had good accuracy with respect to optoelectronic tracking, which functioned as a sensor-based ground truth, specifically the Vicon system (Vicon Motion Systems Ltd, Oxford, GB) [92]. Bandini et al. performed an analogous comparison between a 3D camera (Intel RealSense) and an EMA system (NDI Wave Speech Research System – Waterloo, Ontario, Canada), and demonstrated good accuracy of the camera compared to the Wave [40]. Harkel et al. evaluated the intra- and inter-rater reliability and agreement of the Intel RealSense D415 camera against a gold standard stereophotogrammetry imaging system, 3dMD, in patients with facial palsy and found that the 3D camera was a viable option to be implemented in the context of an objective clinical assessment [94]. Taken together, these previous studies have extensively demonstrated the analytical validity of 3D camera-based facial tracking methods.

Beyond verification and analytical validation, the technology (both 3D camera and deep learning algorithms) must be evaluated in clinical populations in order to fulfill the third step of biomarker development. Previous work has shown promise for detecting orofacial impairments associated with stroke [116], Parkinson's disease [41], [101], and amyotrophic lateral sclerosis (ALS) [42] relative to neurologically healthy controls, based on measures of facial kinematics

(e.g., velocity, acceleration, range of motion, symmetry, etc.). For example, Bandini et al used a small feature set capturing movement symmetry, speed, and range of motion (ROM) of the lips and jaw to automatically differentiate between post stroke patients and controls with 87% accuracy [116]. In another study, a small number of 3D kinematic features of lip movement were used to discriminate patients with ALS from healthy subjects with nearly 90% accuracy [42]. Neumann et al. used a combination of kinematic features extracted from 2D video recordings together with acoustic features extracted from audio data to distinguish patients with ALS at symptomatic versus pre-symptomatic phases of bulbar disease as well as from the control participants [117].

The focus of previous studies was to investigate video-based methods' use for classifying a group of participants into either the healthy or clinical category. The studies predicting the severity of symptoms have been limited, but important in the effort to clinically validate novel assessment technologies. One study showed that kinematics of facial expressions was able to predict the depression severity estimated via the Beck Depression Inventory-II (BDI-II) [105]. In another study, features were extracted from facial expression images of participants in order to automatically differentiate among five levels of pain using AI-based models evaluated its efficacy on two open-source datasets [118]. Additionally, the Unified Parkinson's Disease Rating Scale (UPDRS) motor score was predicted using acoustic features extracted from 5875 voice recordings of 42 patients with PD with the final ensemble model predicting the clinical score with 99.6% accuracy [119]. These promising results highlighted the need for more research to clinically validate automated facial analysis technologies for their use as assessment tools, specifically for predicting the clinical scores associated with severity of orofacial impairments. The ability to predict disease severity using kinematic measures extracted from videos would provide evidence of the clinical validity of the technology and support its use to enable tracking disease progression as well as evaluating the effectiveness of intervention programs (e.g., quantifying functional improvement post-stroke).

The first objective of this study was to establish clinical validity of a video-based system in assessing the severity of the orofacial impairments by automatically extracting kinematic features from video recordings of individuals with orofacial impairments either due to stroke or MND and comparing them with respect to their corresponding clinical scores (e.g., clinical estimates of symmetry/speed/ROM and kinematic measures of symmetry/speed/ROM). The

second objective was to determine whether a combination of kinematic features was able to predict the overall severity of orofacial deficit in each patient group. We hypothesized that the selected kinematic features and corresponding perceptual clinical scores would be significantly correlated and could be used to predict the overall severity of orofacial impairment.

3.2 Methods

3.2.1 Participants

Data from 45 patient participants were included in this study: 19 individuals were diagnosed with motor neuron disease (MND, including ALS= 16; primary lateral sclerosis= 1; Kennedy's disease= 2), and 26 individuals who experienced a stroke (PS). We combined all MND patients into the same group as the manifestation of the disease in the orofacial mechanism was expected to be similar (e.g., resulting primarily in kinematic changes due to primarily muscle weakness). Table 3-1 presents the demographic and clinical representation of the samples. The more in-depth clinical details of participants are provided in Table A-2 and Table A-3 in Appendix A2. All participants were cognitively unimpaired (Montreal Cognitive Assessment - MoCA score ≥ 26 [120] and passed a hearing screening at 25 dB for both ears at 500, 1000, 2000, and 4,000 Hz. The MND group was also evaluated via ALS – Functional Rating Scale – Revised (ALSFRS-R) [121]. This measure is routinely used by clinicians to evaluate bulbar dysfunction and consists of only 3 questions to address changes in speech, swallowing, and salivation that is rated on a four-point ordinal scale with 0 indicating no function and 4 indicating normal function [122]. The study was approved by the Research Ethics boards at the Sunnybrook Research Institute and University Health Network: Toronto Rehabilitation Institute, Toronto, Canada. All participants provided informed consent according to the requirements of the Declaration of Helsinki.

Table 3-1Demographics and clinical information (means and standard deviations) of the two participants groups: Motor Neuron Disease (MND), and Post-Stroke (PS).

Population	N (Female)	Age	Months since diagnosis	ALSFRS – R (bulbar sub score)
MND	19 (9)	62.12 ± 8.24	45.62 ± 73.78	9.87 ± 2.03
PS	26 (9)	66 ± 12.81	22.13 ± 28.32	-

3.2.2 Clinical Assessment

A highly experienced speech-language pathologist specializing in the assessment of adults with neurological diseases (> 10 years of clinical experience) watched the available video recordings and rated them based on clinical measures of ROM, speed, symmetry, variability, and fatigue of facial movements. The rater only knew the diagnosis of the participants prior to performing the assessment and otherwise was blind to all other clinical information (e.g., scores on other clinical assessments, site of onset, disease duration, etc.). Ratings were based on a 5-point Likert scale with 1 indicating normal function and 5 indicating severe dysfunction. ROM, speed, and symmetry scores were used as separate clinical outcomes; they were selected as they constitute salient neuromuscular abnormalities in motor speech disorders [123]. The total score – a sum of all 5 clinical ratings – was calculated and used to represent the overall orofacial deficit severity, with score of 5 indicating normal function and score of 25 indicating the most severe dysfunction (see [73]).

To determine the reliability of the clinician scores, the original examinator (intra-rater reliability) and another trained examinator (a certified SLP; interrater reliability) re-scored videos from 15% of the sample (7 participants chosen randomly; 21 recordings). To assess intra-rater reliability, Intraclass Correlation Coefficient (ICC) estimates and their 95% confidence intervals (CI) were calculated using SPSS statistical package version 23 (SPSS Inc, Chicago, IL) based on a single-rating, absolute-agreement, 2-way mixed-effects model [124]. To assess inter-rater agreement, ICC estimates and their 95% CI were calculated based on a single-rating, consistency-agreement, 2-way random-effects model [124]. A percentage of clinical agreement (percentage of scores within 1-point difference) was also calculated to mitigate the effects of skewed distributions or low variability of the scores in ICC values, if necessary. The intra-rater agreement was strong (ICC = 0.966, CI = 0.908 – 0.987, $p < .001$; [125]). The inter-rater agreement was also excellent (ICC = 0.996, CI = 0.990 -0.998, $p < .001$; [125]). All scores fell within 1-point difference for both intra-rater and inter-rater reliability.

3.2.3 Data collection procedures

Video recordings were obtained while participants were seated in front of an Intel RealSense depth camera (SR300 or D400) at approximately 30-60 cm distance away from the face

at the eye level height. The recordings were performed in a quiet room with a single source uniform light illuminating the participants' face. Each video recording was composed of a pair of color (RGB) and depth videos. Both streams were recorded simultaneously at approximately 50 frames per second at VGA resolution of 640×480 pixels.

Participants were asked to perform a set of standard speech and non-speech tasks commonly used by clinicians during orofacial and motor speech examinations to assess the severity of orofacial symptoms and dysarthria. The subjects were asked to avoid head and body movements and look directly into the camera during recordings. The following set of tasks were chosen for analysis as they captured the orofacial movements of mouth and jaw in both horizontal and vertical directions and were affected in both clinical populations in our previous studies [23], [39]:

- 10 repetitions of the sentence "Buy Bobby a Puppy" at a comfortable speaking rate and loudness (BBP)
- 5 repetitions of maximum jaw opening and closing (OPEN)
- 5 repetitions of pretending to smile with tight lips (SPREAD)

Furthermore, we recorded 10 seconds of participants resting, while they maintained a neutral facial expression (REST) with their teeth together but not clenched. This was done for distance normalization purposes.

A total of 175 video recordings were included in the final analysis: 74 from MND, and 101 from PS. A total of 5 recordings were either missing or excluded. Reasons for exclusion of a video recording for a task were: participant not performing the task (2 cases – in MND group, BBP task), video recording issues, e.g., corruption of either the colour and/or depth stream data of video recordings, or dropping of video frames (2 case – in PS group, BBP task), and failed accurate facial landmark tracking due to poor lighting condition (1 case – in PS group, BBP task). Each of the remaining videos, excluding the data for the REST task, were manually segmented into individual repetitions by a trained research assistant (i.e., 5 segments for the non-speech tasks and 10 segments for the speech task).

3.2.4 Kinematic Data Analysis

Preprocessing

A fine-tuned version of the FAN [80] was used to extract 68 facial landmarks from colour videos in 2 dimensions (2D) pixel values and track them throughout the duration of the videos (see Figure 3-1). The 3D positions of landmarks were then calculated by aligning and sampling the depth images at the same pixel coordinates as the landmark points identified on the color images. The 3D real world coordinates of the landmarks (in meters) were calculated using the intrinsic camera parameters together with aligning the depth and colour streams, thus, obtaining 3D coordinates for each landmark, with X representing the lateral axis, Y representing the vertical axis, and Z representing the frontal axis.

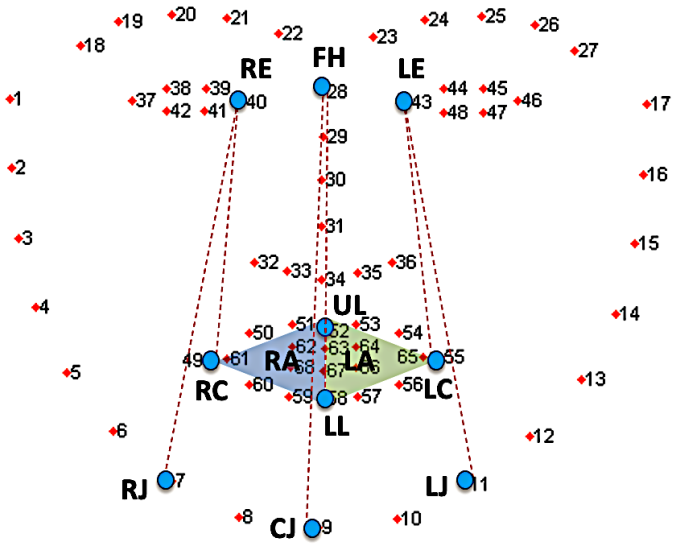

Figure 3-1 Facial landmarks obtained from the fine-tuned FAN. The landmarks used for extracting kinematic features are named and highlighted in light blue dots. The red dashed lines are the kinematic displacement measures extracted from specific facial landmark pairs.

It has been previously shown that the state-of-the-art deep neural facial landmark detection models do not perform well when used on faces of older adults or patients with orofacial deficits [73], [83]. To address this issue, the model was previously fine-tuned using 4340 images extracted from video recordings of individuals from clinical populations as well as healthy older adults, some of which were of a subset of the individuals in the present study, performing orofacial tasks [126]. The normalized Root Mean Squared Error (NRMSE) of the fine-tuned model was reported

to be 4.7 ± 0.7 %, 6.2 ± 1.7 %, and 5.7± 1.1 % for healthy control, post-stroke group, and ALS group, respectively [126]. The fine-tuned FAN model has been previously used to successfully detect oromotor impairments associated with various neurological disorders [73], [83]. To check the model's accuracy, a video file was generated of participants performing each task with the location of 2D landmarks superimposed on their faces. Each video was viewed to ensure adequate detection and tracking of facial landmarks of interest.

Out of 68 facial landmarks, 10 landmarks from lips and jaw were considered in this study. They included: Forehead (FH), left medial canthus of eye (LE), right medial canthus of eyes (RE), upper vermillion lip border at mouth centre (UL), lower vermillion lip border (LL) at mouth centre, left oral commissure (LC), right oral commissure (RC), centre jawline (CJ), left jawline (LJ), and right jawline (RJ). To remove tracking noise and jitter, the trajectories of the landmarks of the interests were lowpass filtered with an 8-pole Butterworth filter with cut-off frequency of 12 Hz [73], [127].

Kinematic Features

Similar to previous studies, facial landmarks were used to derive clinically interpretable and relevant kinematic features [42], [84]. These features are summarized in Table 3-2. Features were selected to correspond to clinical judgements of ROM, speed and symmetry by the speech language pathologist. A total of 20 kinematic features were linked to ROM, 24 to speed, and 9 to symmetry. All the features have been used in previous studies [23], [101].

ROM Features

The 3D Euclidean distances between a set of facial landmarks were calculated to establish lip opening, UL – LL; lip spread, RC – LC; movement of LL with respect to the FH reference point; movement of RC with respect to RE; and movement of LC with respect to LE. In addition, the total movement of the mouth was modelled by a sum of the areas made by two triangles defined by lip landmarks {RC, UL, LL} and {LC, UL, LL}. Lip spread as well as other RC/LC measures were established only for SPREAD, where horizontal movements were predominant. Since the jaw rarely moves during SPREAD, none of the jaw-based features were considered for this task. All the calculated distance features were normalized with respect to the corresponding values

calculated in the REST condition. After this normalization, ROM was determined as the maximum value of each of these distances/areas, as well as the cumulative sum path length. This was done for each repetition.

Speed Features

The first derivative of the above path distances with respect to time were calculated to represent the velocity of the lips and jaw movements. The maximum, minimum, and average values were considered.

Symmetry Features

The symmetry of movement was characterized by (1) comparing the motion of the left and right corners of the mouth with respect to the reference points LE and RE and (2) comparing the areas on the right and left triangles defined by {RC, UL, LL} and {LC, UL, LL}, respectively. The absolute differences, ratios, and concordance correlation coefficient (CCC [128]; as a measure of agreement of two signals) between the computed values of the distances and mouth areas on the right and left sides were calculated. Smaller CCC values corresponded to asymmetry during facial movements. Similarly, the larger deviation of right/left ratios away from 1 indicated more asymmetrical facial movements.

Table 3-2 Description of kinematic features extracted

Type/ Clinical Interpret-ation	**Oro-facial Structure**	**Feature Description**	**Feature Label**	**Tasks**
Range of Motion (ROM)	Lips	Distance between UL and LL, Max and cumulative sum of path values	Open-Max Open-Path	BBP, OPEN
		Distance between RC and LC, Max and cumulative sum of path values	Spread-Max Spread-Path	SPREAD
		Distance between LL and FH, Max and cumulative sum of path values	LL-Max LL-Path	BBP, OPEN
		Distance between RC and RE, Max and cumulative sum of path values	RC-Max RC-Path	SPREAD
		Distance between LC and RE, Max, and cumulative sum of path values	LC-Max LC-Path	SPREAD
		Area of the mouth, sum of two triangles made by {RC, UL, LL} and {LC, UL, LL}, Max, average	tA-Max tA-Avg	BBP, OPEN
	Jaw	Distance between RJ and RE, Max Cumulative sum RJ-RE path distance	RJ-Max RJ-Path	BBP, OPEN
		Distance between LJ and LE, Max	LJ-Max	BBP, OPEN

		Cumulative sum LJ-LE path distance	LJ-Path	
		Distance between CJ and FE, Max Cumulative sum CJ-FE path distance	CJ-Max CJ-Path	BBP, OPEN
Speed of Motion	Lips & Jaw	Velocity calculated as the first derivative of the above defined distances with respect to time — Max, Average, and Min values	vOpen-Max, -Min, -Avg vSpread-Max, -Min, -Avg vLL-Max, -Min, -Avg vRC-Max, -Min, -Avg vLC-Max, -Min, -Avg vRJ-Max, -Min, -Avg vLJ-Max, -Min, -Avg vCJ-Max, -Min, -Avg	OPEN, SPREAD, BBP as above
Symmetry	Lips	Comparison of the two distances defined by landmarks {RC, RE} and {LC, LE} — Absolute difference, ratio, and concordance correlation coefficient (CCC)	RCLC-dif RCLC-R-Avg RCLC-CCC	SPREAD
		Comparison of the two triangle areas made by {RC,UL,LL} and {LC,UL,LL} as vertices. Absolute difference, ratio of the range={Max - Min}, average, and CCC of the right area with respect to the left area	RALA-difference RALA-R-Range RALA-R-Avg RALA-CCC	BBP, OPEN

3.2.5 Statistical Analyses

The features described above were calculated for individual repetitions of all tasks. Prior to further analyses, outliers were removed from the data using the inter-quartile range (IQR) method [129]. In this method, the IQR of the repetitions were calculated, then multiplied by 2.5, and added/subtracted from the third/first quartiles. The values per repetition of each task that fell outside this range were considered outliers and were eliminated from the set. In total, 4.7% of the repetitions were discarded as outliers (n=51). The arithmetic mean of kinematic feature values were calculated cross the remaining repetitions per subject, per task. All subsequent analyses were conducted separately for each task (OPEN, SPREAD, BBP) and group (MND vs. post-stroke).

The Spearman correlation coefficients were calculated between each kinematic feature and their corresponding perceptual clinical scores. The Spearman rho (r) is used with ordinal data and measures the degree of monotonic relationship between two variables; here, in accordance with accepted guidelines, values between 0 to ±0.20 were considered as negligible, ±0.21 to ±0.40 as weak, ±0.41 to ±0.60 as moderate, ±0.61 to ±0.80 as strong, and ±0.81 to ±1.00 as very strong [130]. The significant level for each correlation was set to 0.05 to reflect that these correlations were primarily used as a screening method for subsequent regression analyses.

Kinematic features with statistically significant correlations were included in the subsequent multivariable linear regression models with the total orofacial clinical score as the outcome. The Variance Inflation Factor (VIF) with threshold of greater than 10 was used to assess the collinearity between covariates [131]. If the VIF for the variables were higher than the set threshold, the features with highest VIF were eliminated iteratively until the remaining variables all had VIF values less than 10. The normality of residuals was tested by plotting the Q-Q norm of the residuals. The linearity of fitted regression models was also evaluated by plotting the residual values against the kinematic features data. The adjusted R^2, adjusted for the number of predictors, was reported for the final fitted regression models.

3.3 Results

3.3.1 Clinical Scores

Figure 3-2 shows the distribution of the severity of the clinical scores for each orofacial task and participant group. Notably, the variation of the clinical scores was task dependent; a larger spread of all scores was noted for the SPREAD task in both groups and to some extent in BBP, particularly for the MND group.

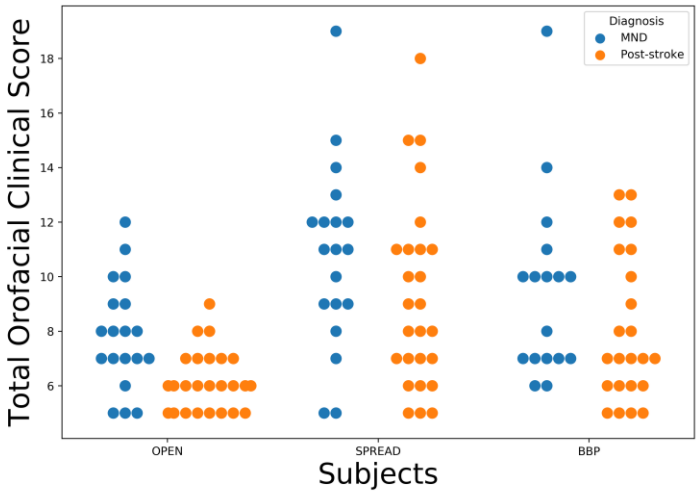

Figure 3-2 Distribution of total clinical scores, blue dots represent the MND participants while the orange dots represent the post-stroke participants

3.3.2 Correlation analyses between clinical scores and kinematic measures

Tables 3-3, 3-4, and 3-5 present the Spearman correlation coefficients between the extracted 3D orofacial kinematic features and their corresponding perceptual scores of ROM, speed, and symmetry, respectively, for each task (OPEN, SPREAD, and BBP) separately.

In the OPEN task, kinematic features of ROM and speed (7 features in total) showed significant correlations with the corresponding clinical scores only in the MND group, while none of the features were significant for the post-stroke group. A total of 16 kinematic measures (8 for MND and 8 for post-stroke) showed significant correlations with clinical scores with the majority of measures in the speed category. For the BBP task, only one ROM feature had a statistically significant correlation with the clinical measure of ROM in the MND group; at the same time 15 kinematic features, mostly in the ROM and symmetry category, were statistically significant in the post-stroke group.

*Table 3-3 Clinical features and their correlations to their corresponding clinical scores for OPEN task, *significant at $p < 0.05$*

Clinical Score	**Kinematic Feature**	**MND**	**Post-Stroke**
		Spearman r	**Spearman r**
Range of Motion (ROM)	Open_Max	-0.56*	-0.04
	Open_Path	-0.37	0.04
	LL_Max	-0.48*	-0.10
	LL_Path	-0.27	-0.04
	RJ_Max	-0.50*	-0.22
	RJ_Path	-0.21	-0.12
	LJ_Max	-0.41	-0.33
	LJ_Path	-0.18	-0.25
	CJ_Max	-0.44	-0.37
	CJ_Path	-0.15	-0.22
	tA_Max	-0.19	-0.18
	tA_Avg	-0.44	-0.14
Speed	vOpen_Max	-0.54*	-0.08
	vOpen_Min	0.58*	0.00
	vOpen_Avg	0.42	0.15
	vLL_Max	-0.55*	0.02
	vLL_Min	0.72*	0.02
	vLL_Avg	-0.09	0.00
	vRJ_Max	-0.25	-0.35
	vRJ_Min	0.36	0.20

	vRJ_Avg	-0.03	-0.21
	vLJ_Max	-0.29	-0.06
	vLJ_Min	0.37	0.12
	vLJ_Avg	-0.45	-0.22
	vCJ_Max	-0.38	-0.20
	vCJ_Min	0.39	0.11
	vCJ_Avg	0.06	-0.24
Symmetry	RCLC_difference	-0.17	0.20
	RCLC_CCC	-0.09	-0.13
	RCLC_R_Avg	-0.36	-0.21
	RALA_ difference	-0.20	-0.25
	RALA_CCC	0.10	0.03
	RALA_R_Avg	0.02	-0.16
	RALA_R_Range	-0.12	-0.03

*Table 3-4 Clinical features and their correlations to their corresponding clinical scores for SPREAD task, *significant at $p < 0.05$*

Clinical Score	**Kinematic Feature**	**MND**	**Post-Stroke**
		Spearman r	**Spearman r**
Range of Motion (ROM)	Spread_Max	-0.57*	-0.46*
	Spread_Path	-0.19	-0.13
	RC_Max	0.31	-0.29
	RC_Path	0.17	0.13
	LC_Max	0.32	-0.12
	LC_Path	0.33	0.16
Speed	vSpread_Max	-0.48*	-0.52*
	vSpread_Min	0.66*	0.50*
	vSpread_Avg	-0.20	-0.27
	vRC_Max	-0.58*	-0.28
	vRC_Min	0.40*	0.41*
	vRC_Avg	0.09	0.15
	vLC_Max	-0.58*	-0.36
	vLC_Min	0.48*	0.40*
	vLC_Avg	0.19	0.23
Symmetry	RCLC_difference	0.21	0.29
	RCLC_CCC	-0.58*	-0.54*
	RCLC_R_Avg	0.44	0.29
	RALA_ difference	0.03	0.13
	RALA_CCC	-0.41	-0.11
	RALA_R_Avg	-0.09	-0.46*
	RALA_R_Range	0.00	-0.42*

*Table 3-5 Clinical features and their correlations to their corresponding clinical scores for BBP task, *significant at p < 0.05*

Clinical Score	Kinematic Feature	MND	Post-Stroke
		Spearman r	Spearman r
Range of Motion (ROM)	Open_Max	-0.10	0.40*
	Open_Path	0.14	0.39
	LL_Max	-0.05	0.41*
	LL_Path	0.24	0.46*
	RJ_Max	0.42	0.62*
	RJ_Path	0.50	0.51*
	LJ_Max	-0.65*	0.33
	LJ_Path	-0.47	0.44*
	CJ_Max	-0.22	0.47*
	CJ_Path	0.17	0.54*
	tA_Max	-0.04	0.44*
	tA_Avg	-0.23	0.28
Speed	vOpen_Max	-0.26	0.46*
	vOpen_Min	0.19	-0.32
	vOpen_Avg	-0.07	-0.06
	vLL_Max	-0.15	0.44*
	vLL_Min	0.10	-0.23
	vLL_Avg	-0.03	-0.07
	vRJ_Max	0.45	0.30
	vRJ_Min	-0.49	-0.20
	vRJ_Avg	0.26	-0.01
	vLJ_Max	-0.47	0.47*
	vLJ_Min	0.40	-0.41
	vLJ_Avg	0.33	0.21
	vCJ_Max	-0.17	0.36
	vCJ_Min	0.01	-0.23
	vCJ_Avg	0.28	-0.13
Symmetry	RCLC_difference	0.13	0.66*
	RCLC_CCC	-0.20	-0.33
	RCLC_R_Avg	0.34	0.36
	RALA_ difference	0.07	0.18
	RALA_CCC	0.06	0.02
	RALA_R_Avg	0.09	-0.74*
	RALA_R_Range	-0.05	-0.56*

3.3.3. Predicting the overall severity of orofacial impairment using kinematic measures

The VIF analysis eliminated one kinematic feature for OPEN (vOpen_Max), and two features for SPREAD (e.g., vRC_Max and vLC_Max) in the MND dataset. In the post-stroke case, a single kinematic feature (vSpread_Max) was eliminated for SPREAD, and seven features were eliminated for BBP (Open_max, LL_Max, LL_Path, RJ_Max, RJ_Path, LJ_Path, and vLL_Max).

Table 3-6 presents the regression coefficients that predicted the total clinical scores based on kinematic features for the MND group. For OPEN, kinematic features predicted the overall clinical score with adjusted R2 value of 0.50 (p-value <0.01), and five out of six kinematic predictors were statistically significant. For SPREAD, the adjusted R2 was 0.35 (p-value <0.01); for BBP, it was 0.11 (p-value ~0.09). No individual features were significant for the SPREAD or BBP models.

Table 3-6 Regression coefficients, standard error, t-values, p-values, and 95% confidence interval for multivariate regression models for the MND group

	Standardized Regression Coefficient	**Std. Error**	**t-value**	**95% Confidence Interval**
Task / Kinematic Features	OPEN			
Open_Max	-2.26	0.92	-2.46*	[-4.25, -0.26]
LL_Max	2.23	1.04	2.14	[-0.04, 4.51]
RJ_Max	-1.02	0.44	-2.34*	[-1.97, -0.07]
vOpen_Max	-2.08	0.94	-2.21*	[-4.14, -0.03]
vLL_Min	1.80	0.58	3.10*	[0.53, 3.06]
vLL_Max	2.69	1.10	2.45*	[0.30, 5.09]
Task / Kinematic Features	SPREAD			
Spread_Max	-0.96	0.81	-1.19	[-2.72, 0.79]
vSpread_Max	1.45	1.49	0.97	[-1.79, 4.69]
vSpread_Min	1.80	1.09	1.66	[-0.57, 4.17]
vRC_Min	-1.26	1.75	-0.72	[-5.06, 2.54]
vLC_Min	2.14	1.81	1.18	[-1.81, 6.09]
RCLC_CCC	-1.11	0.85	-1.13	[-2.95, 0.74]
Task / Kinematic Features	BBP			

LJ_Max	-0.78	0.48	-1.64	[-1.81, 0.25]

*Significant at p < 0.05, **significant at p < 0.01

Table 3-7 presents regression results for the post-stroke group. For SPREAD, the adjusted R2 value was 0.57 (p-value <0.01) and significant predictors included: vSpread_Min, vRC_Min, and RCLC_CCC. For BBP, the adjusted R2 value was 0.54 (p-value <0.01), and significant predictors were tA_Max and vO_Max. A regression model was not built for the OPEN task as none of the features had statistically significant correlations with their individual corresponding clinical scores.

Table 3-7 Regression coefficients, standard error, t-values, p-values, and 95% confidence interval for multivariate regression models for the PS group

	Standardized Regression Coefficient	**Std. Error**	**t-value**	**95% Confidence Interval**
Task / Kinematic Features	SPREAD			
Spread_Max	0.53	0.63	0.84	[-0.80, 1.86]
vSpread_Min	2.49	0.98	2.54*	[0.42, 4.56]
vRC_Min	-2.55	1.08	-2.36*	[-4.84, -0.27]
vLC_Min	0.87	0.86	1.02	[-0.93, 2.68]
RCLC_CCC	-1.53	0.52	-2.91**	[-2.63, -0.42]
RALA_R_AVG	-1.26	0.60	-2.11	[-2.53, 0.00]
RALA_R_RANGE	-0.18	0.69	-0.26	[-1.65, 1.28]
Task / Kinematic Features	BBP			
CJ_Max	1.06	0.96	1.10	[-1.01, 3.13]
CJ_PATH	-0.07	1.06	-0.07	[-2.32, 2.18]
tA_Max	-2.78	1.26	-2.21*	[-5.47, -0.08]
vO_MAX	2.12	0.69	3.06**	[0.63, 3.60]
vLJ_MAX	0.20	0.60	0.33	[-1.10, 1.49]
RCLC_diff	0.74	1.14	0.65	[-1.69, 3.17]
RALA_R_AVG	-1.25	0.68	-1.84	[-2.72, 0.21]
RALA_R_RANGE	-1.13	0.78	-1.44	[-2.81, 0.55]

*Significant at p < 0.05, **significant at p < 0.01

3.4 Discussion

In this exploratory study, we investigated clinical validity of a 3D camera and vision-based facial tracking algorithm as an objective assessment tool for patients with neurological disorders – MND and PS. A relatively small, clinically interpretable lip and jaw kinematic feature set was delineated from the outset; the features were chosen to conceptually represent the measures used by clinicians to assess orofacial deficits during the oromotor examinations of non-speech and speech tasks. The study revealed that subsets of kinematic features were significantly correlated with their corresponding clinical scores for each clinical group. The sets of statistically significant features that were able to explain a significant portion of data variability in the overall severity scores in most models were unique for each of the clinical subgroups. Overall, the findings demonstrated the potential of automated video-based systems for extraction of kinematic features in the context of orofacial neurological assessment and established their initial validity as clinical tools.

3.4.1 Facial kinematics in MND and PS

In total, there were 16 statistically significant correlations (out of 50, 32%) across all three tasks for the MND group. In the PS group, 23 individual correlations (8 for SPREAD and 15 for BBP) of kinematic features with clinical scores were significant (46%). Although less than 50% of correlations were significant, and those that were significant ranged between 0.40 and 0.74 (moderate to good), the results were overall encouraging. Smaller correlations and model fits have been expected based on previous literature exploring relationships between kinematics and other measurement domains, including perceptual and acoustic (see Lee et al., 2018; Rong et al., 2012; Teplansky et al., 2022).

However, the findings of this study aligned strongly with those of previous work with regards to facial kinematics and oromotor phenomena in both MND and PS groups. In the MND group, our findings were consistent with previous literature describing abnormal jaw and lip movements [42], [127], [135], [136]. MND has been characterized by weakness in the orofacial musculature [137] and, expectedly, ROM and speed measures were noted to be substantially more affected by MND than symmetry (see Shellikeri et al., 2016). As a result, we observed significant

correlations between the clinical scores indicating the impairment severity and kinematic features primarily related to ROM and speed.

In the PS group, facial paresis, resulting in facial asymmetry with a unilateral muscle impairment of lower half of the face, causing asymmetric smile and other facial gestures, is one of the most common impairments following a stroke [116], [139]. As anticipated, the symmetry features played a prominent role in the PS group, and our finding showed significant correlations between kinematically assessed symmetry and the corresponding symmetry clinical scores. In both clinical groups, the agreement between clinical characteristics sought by experts to describe the unique clinical representation of orofacial deficits and corresponding kinematic features supported the overall impression of clinical validity of the video-based movement assessment.

3.4.2 Task differences

Differences also emerged in this study between tasks. In the MND group, kinematic measures of ROM and speed were linked to the corresponding clinical scores in non-speech OPEN and SPREAD tasks, but not in the speech task-BBP. The findings for non-speech tasks were anticipated based on past research [42]. Both tasks are classified as maximum performance tasks, yet clinically they serve different purposes. While OPEN primarily is used to judge the integrity of jaw depressor/elevator muscles, SPREAD is used to judge the integrity of intrinsic facial muscles. Previously, kinematic features extracted from lower lip and jaw in SPREAD and OPEN were used to differentiate patients with ALS from healthy control participants with 83% and 72% accuracy, respectively [42]. The findings regarding the speech task were somewhat unexpected, as previous studies had shown BBP to be highly sensitive to the presence of the motor neuron disease [23], [127]. Current results for BBP may be due to the challenges associated with clinical (visuo-perceptual) assessment of speech versus non-speech movements by SLPs. BBP is a relatively fast task and as such it might be more challenging for clinicians to rank its clinical characteristics (ROM, speed, and symmetry) on a 5-point scale. Further, only 2D colour video recordings (no audio) were viewed by the expert for clinical judgement of BBP, which might have resulted in a loss of some information (e.g., rate/slowness) essential for forming the overall clinical impression. Limited sample size and high variability (due to factors such as heterogeneity of disease presentation) may have also contributed to these results.

In the PS group, results for the vertical opening (OPEN) contrasted with the horizontal opening (SPREAD) and speech (BBP) task. None of the kinematic features were correlated with clinical scores in OPEN, while sets of features were correlated with the clinical scores in SPREAD and BBP. The lack of correlation among features and the clinical scores in OPEN may be due to the task being mainly used to assess the integrity of the jaw (i.e., cranial nerve V) and not facial muscles, which might be differentially affected post-stroke [116]. The SPREAD task showed more notable effect as it intended to reveal the integrity of facial muscles (i.e., cranial nerve VII). Although relatively mild, this type of impairment predominated in our PS sample. Facial features during BBP also revealed a link with the clinical symmetry score in these patients. Similar results for SPREAD and BBP with respect to their sensitivity to orofacial impairment post-stroke were reported in previous work (Bandini, Green, Richburg, et al., 2018). Overall, with the exception of BBP task in the MND group, task effects observed clinically and kinematically in this study demonstrated results consistent with previous research and, by extension, supported the potential for clinical validity of the tested assessment method.

3.4.3 Modeling overall orofacial impairment severity

A combination of kinematic features was able to predict the overall orofacial impairment severity score in each clinical group and some tasks. This was important to examine as the overall severity score combined clinical impressions across different specific observations that might be difficult to rank separately and individually. In the MND group, significant models were noted for OPEN and SPREAD. All but one individual variable (LL_Max) contributed to the overall model in OPEN explaining a substantial degree of variability of the total clinical score. The value of the coefficients generally suggested that as the severity increased, the values of the ROM and speed of the lips and jaw were reduced. The SPREAD model was significant, but none of the kinematic variables significantly contributed to explaining the model's overall variance. This may suggest that the SPREAD was not an ideal task for feature extraction in this group, as previous research suggested that the greatest impairment in MND affects the vertical movement plane of the jaw and lips [136].

In the PS group, in contrast with MND, most of the features were significant in the regression model for SPREAD, with kinematic features explaining over half of the variance in the

data. The results of this study are promising as the variability of the kinematic measures within and in between subjects in the study were considerable. The value of the coefficients revealed that as the severity increased, the lip movement became more reduced and asymmetrical. In BBP task, total mouth area and the maximum mouth opening velocity were the two statistically significant features with the model being able to explain substantial amount of the variance in the data. These features align with clinicians' judgement as they assess the overall mouth movement and speed of movement during a fast task such as BBP. Overall, the features related to symmetrical movement of the mouth demonstrated to be of the most importance aligned with the clinical perceptual assessment of this clinical group, further strengthening the validity of this method for future automatic objective assessment.

3.4.4 Limitations and future directions

This work had several limitations. The primary limitation was a relatively small sample size and differences in sample sizes of the two clinical groups, limiting our ability to fully compare group results. Another limiting factor was a relatively small severity range in both clinical samples. We capitalized on the greater variability available in our samples based on specific tasks instead. In addition, the kinematic features extracted were averaged across the repetitions for each subject per task and thus variability of the facial movements during each repetition of the task were not considered. Future work will extend on the results obtained in the study, focus on gathering data from a larger/ balanced groups of participants with more severe orofacial impairments as well as developing predictive models using timeseries of facial landmarks instead of averaging kinematic feature values across repetitions. This work will be essential to develop and integrate modern digital technologies into clinical care as well as outcome measures for clinical trials [88].

The present work had limitations to acknowledge and to address in future studies. The primary limitation was a relatively small sample size and differences in sample sizes of the two clinical groups, limiting our ability to fully compare group results. Small sample sizes remain an issue for many clinicals studies pre and during the pandemic, which expedited the transition of this type of work into the domain of remote data collections/ assessment, for which the current study results are of a particular importance. Further, the intra- and inter-rater reliability was performed on a small sub-sample (15% of the sample). Although it is relatively standard to evaluate

approximately that amount of data to establish reliability, the sample there may have been small, introducing the possibility of perceptual bias. Nevertheless, we observed small confidence intervals in the reliability analyses, which supported the adequacy of this method for its present purposes (i.e., evaluating the performance of raters at a sample level, rather than at a participant level).

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Future work will extend the results obtained in this study, while focusing on gathering data from a larger/balanced groups of participants with more severe orofacial impairments as well as developing predictive models using timeseries of facial landmarks instead of averaging kinematic feature values across repetitions. This work will be essential to develop and integrate modern digital technologies into clinical care as well as outcome measures for clinical trials [88]. This study laid the foundation for larger scale studies with larger sample size across different cohorts to ensure clinical validity of the proposed technology as well as generalizability of the results.

3.5 Conclusion

The result of this study showed that the automatic extraction of facial kinematic features from facial video recordings of patients performing standard orofacial assessments is a viable solution for an objective and continuous assessment of facial function. Here we demonstrated encouraging support for the validity of our automated assessment method for detecting the severity of kinematic facial impairments. Development of such systems and their validation with larger datasets holds great promise for the development of clinical tools that can help clinicians objectively assess oromotor and speech disorder and track progression of neurological diseases.

Chapter 4 – General Discussion

Need for Automatic Objective Orofacial Assessments

The goal of this thesis was to validate the use of 3D camera together with deep facial landmark tracking algorithms (the FAN and the fine-tuned FAN) to extract orofacial features aimed at providing objective clinically interpretable measures/ biomarkers to guide clinical decision making in diagnosis, prognosis, and treatment management of neurological disorders. Neurological disorders are heterogenous in their clinical presentation, making them challenging to diagnose [140]. Furthermore, the lack of standards to follow in the clinical setting, and subjectivity of the existing methods used by experts can result in delays in obtaining diagnosis and delayed interventions and treatments. As a result, the clinical management and patient experience is negatively impacted [141].

Studies have shown that subtle changes in speech and orofacial movements may be early signs of disease [8]. The use of markerless video-based technologies can provide clinicians with objective automatic analysis of facial movements and speech patterns that are non-invasive, cost effective, and possibly sensitive to subtle changes that are not noticed by the clinicians [127]. In fact, video-based technologies have been used in orofacial assessments to differentiate between groups of individuals with various neurological diseases and healthy control participants [39], [40], [43], [106], [127], as well as differentiating among various stages of neurological diseases [117], [127]. Therefore, development of such technologies can help support clinicians in the assessment, diagnosis, prognosis, and treatment planning processes.

AI-Vision based Automatic Assessment

Machine learning and deep learning techniques are the dominant methods used in automatically analyzing videos in the field of computer vision. New architectures with higher accuracy are being actively developed in this fast-paced research community. While the current top performing algorithms may report high accuracy relative to manual human annotators on the dataset that they were trained on, they perform worse on datasets that they were not trained on [142]. The other limitation of such algorithms is their less-than-optimal performance in cases of occlusions such as glasses, facial hair, extreme poses, or diverse environmental conditions such as

lighting [45]. These events are extremely difficult to control, and they can compromise the data quality. In our work, we opt to perform data collection in a semi-structured environment, when a research assistant or clinician instructs a patient/ participant to avoid occlusion and maintain optimized parameters for the duration of the recording session.

In addition, most of these models are supervised methods that are trained on large amount of labelled data that may be inherently biased. In particular, landmark detection and tracking algorithms are mainly trained on the online available pictures of faces, which mostly consist of young subjects with light skin tone [143]. A thorough analysis of available datasets have shown that these corpuses are indeed sex/ gender, race, and age imbalanced [142]. Therefore, as expected, off-the-shelf models are often biased towards older participants, darker skin tone, and patient groups [86], [102], [144]. Training such algorithms on clinical data alone is not practical and unrealistic as the new state-of-the-art deep learning algorithms require large corpus of training data (in the order of thousands to millions of samples). Since most clinical studies have relatively small sample size (~10-150 participants per class) and strict restrictions on data sharing broadly among the research community, fine-tuning based on these samples remains challenging. Nevertheless, technical solutions can be found to address some of the limitations and the inherent biases in these algorithms. Guarin et al. have shown that using examples from only 40 subjects in a controlled environment may be sufficient to improve the performance of the FAN model that was previously shown to be biased towards the older and/or clinical participants [145]. Fine-tuning is one of the effective methods of de-biasing existing state-of-the-art AI models [145]. Taati et al have also successfully applied this method to algorithms for facial tracking landmarks and have shown improved performance on tracking of facial features of older adults with dementia [102].

AI-based applications should be evaluated for their validity prior to their adoption. This work used the well-established V3 framework [89] for the validation of digital biomarkers to evaluate the validity of the proposed video-based technology. As the technical verification of the used technology was previously established [34], [91], this work aimed to further validate this technology both analytically, relative to a technical gold standard, and clinically, relative to an assessment by an expert clinician. Ultimately, robust clinically verified AI-based models need to be externally validated on a clinical cohort that adequately represents the target patient population from various collection sites in order to reduce the risks of overfitting and biases [146].

Analytical Validation

In the first study of this thesis, the proposed technology was analytically validated relative to the gold standard highly precise and accurate technology – EMA. It was a pilot study that involved data collection from three healthy adults performing three standard assessment tasks (OPEN, SPREAD, BBP) at varying conditions (e.g. fast, slow, normal, and asymmetrical). The original intent was to expand the data set to more participants; however, the data collection was impacted by the COVID pandemic, and the study had to be stopped due to the inability to move to a remote data collection mode. This study is currently in progress – a data collection for 20 participants has been completed and will be analysed elsewhere.

The kinematic features extracted in the pilot study, namely ranges of motion/ speeds etc. associated with mouth opening and lips spreading during speech and non-speech tasks were compared between the two technologies using the zero-lag cross-correlations and Spearman correlations. The zero-lag cross correlation measure showed strong agreement between the two methods when comparing the entire kinematic signal per repetition. However, Spearman correlations of the calculated averages and maximum values of kinematic features showed mostly moderate correlations. Two conclusions were drawn from the results of this preliminary study. First, the 30 frames rate per second of the video camera can be a limiting factor since fast facial movements may not be captured. Ouni et al. recommended a minimum of 50 frames per second to ensure capturing all the critical information for fast movements such as speech [92]. Second, the jitteriness of the landmarks in the video resulted in “noisy” distance signals and high variability of kinematic values extracted. In addition, this study demonstrated that there was high variability in the range of orofacial kinematics of orofacial movements performed by participants both within and in-between subjects. The high variance needs to be considered before making conclusions regarding the results of studies using such technologies. Finally, the strengths of the correlations were also dependant on the participants. The highest correlations were obtained on the participant with the lightest skin tone, which is aligned with the findings of previous studies on the inherent biases of deep facial landmark detection algorithms [86], [102], [106], [144], [145].

This study also revealed some of the challenges and limitations of the proposed system. The RealSense video camera sometimes dropped video frames, which resulted in overall lower

video quality. A new recording software was developed in the lab to address this problem and inform the researchers if a specific recording needed to be repeated, based on the number of dropped samples and overall low recording quality. The off-the-shelf FAN model used in this study showed high sensitivity to environmental factors such as variation in lighting as well as skin tone of participants. As such, fine-tuning procedures were established to improve FAN's performance and were used in the clinical validation study and elsewhere in our research.

The main limitation of this study was the small sample size (n=3). Ongoing work focuses on full sample (n=17) evaluation of landmark detection and tracking algorithm on the data collected from EMA, 3D RealSense camera, and a laptop webcam [147]. The result of this study will be reported elsewhere and will include measures of inter-method agreement, test-retest consistency, and reliability of such technologies for performing automatic objective orofacial assessments.

Clinical Validation

In the second study, the proposed technology was clinically validated in the MND and Post-stroke groups. Participants performed the same tasks as first study (OPEN, SPREAD, BBP) while being recorded in a controlled lab environment with a 3D camera. In this study, a fine-tuned model of the landmark detection and tracking algorithm on clinical data [85], [145] was used to automatically extract clinically relevant kinematic features from lips and jaw movements. A set of features in each clinical group showed moderate to good correlation with the perceptual clinical scores given by the expert clinician. The statistically significant features within each task and clinical subgroup were representative of clinical characteristics anticipated by the group. The regression models built by the significant features were able to explain a good amount of variability within the data. High within and between subject variability was also observed in the extracted kinematic features in each of the categories of the severity scale, which highlights the challenges associated with heterogeneity of clinical characteristics and presentations across clinical populations. A recent longitudinal study of post-stroke participants established a high internal consistency and reliability of such technologies with the ability to identify individual and task-dependent differences over time [148].

In this study, we only focused on three out of the five available clinical scores; namely, ROM, speed, and symmetry. There were no objective kinematic measures defined/extracted for the two remaining scores of variability and fatigue. However, these measures are also important in characterizing neurological diseases. For example, individuals with ALS have reported fatigue as the most prevalent symptom occurring in 83% - 90% of patients compared with 20% in healthy control group [149]. Future studies should define objective and clinically interpretable measures for fatigue and variability sub-scores and include them for the full analysis. An example of a clinically interpretable objective measure to consider for fatigue is the rate of change in a given kinematic feature over consecutive repetitions of a task. It is expected that larger clinical scores of fatigue would translate to larger differences between kinematic features from early repetitions to final repetitions. Similarly, one can examine the standard deviation of a given kinematic feature over multiple repetitions as a potential marker of the variability sub-score.

Finally, this study provided some insights on the usefulness of different assessment tasks for quantifying different aspects of orofacial impairments. For example, SPREAD is mainly used to assess symmetry of the horizontal lip motion, while OPEN assesses the vertical symmetry as well as the range of motion of the jaw and lips. Understanding the clinical importance and interpretation from each assessment task can help understand and validate such clinical tools as long as the automatic kinematic features extracted remain clinically interpretable and relevant.

This study showed that predicting the severity of clinical rating scales is challenging and may be limited by 1) the small sample size, and 2) the skewed distribution of the data across severity scale. Future studies should focus on gathering data from participants with more severe orofacial impairment in order to balance the dataset and have representative data in all the categories. Overall, this study provided evidence for validity of such technologies as objective assessment tools for orofacial movements.

Conclusion

The overall goal of this project was part of an effort to help the translation of novel video-based technologies capable of providing objective kinematic measures into a useful clinical tool allowing for more accurate assessment of orofacial deficits within neurological disorder populations. Adopting the V3 framework for validation of digital biomarkers [89], the two studies

were performed on healthy, MND, and post-stroke participant groups to analytically and clinically validated such technologies for objective kinematic assessment of orofacial movements. measure provided by EMA. The combined results indicated that such technology can be used to automatically extract clinically meaningful and objective features that clinicians can incorporate into their workflow to make more informed and data-driven decisions.

Limitations and Future Directions

Upgrading the facial detection and tracking algorithm

In present work, FAN [80] and fine-tuned FAN [85] models were used for facial landmark detection and tracking purposes. FAN is a relatively old model (proposed in 2017) in the domain of deep facial tracking architectures, and lacks testing on widespread face landmark datasets [61]. Current state-of-the-art models have reported higher performance and have been trained on larger and more diverse datasets than FAN. For example, PropagationNet, ADNet, and SubpixelHeatmap are more recent architectures that have achieved top performance on different datasets [61]. They are all based on the similar heatmap architecture as FAN and use Hourglass for their backbone with their own unique improvement; PropagationNet introduced a new face boundary attention module, ADNet separates normal and tangent errors, while SubpixelHeatmap focuses on reducing heatmap quantization error [61]. The higher performance of these algorithms on datasets used for training and testing FAN as well as more challenging datasets [61] suggest their superior performance and higher facial landmark detection and tracking on the clinical data collected for these studies. Using the methods in [85] and clinical orofacial datasets such as [73], these models can be further fine-tuned to more accurately track the facial features in clinical populations. Increasing the performance of landmark detection and tracking algorithms will directly translate to extracting more accurate kinematic features and thus development of more reliable and accurate tool for assessing orofacial movements.

Using all the spatial and temporal data from facial landmarks

The current studies used the values of kinematic features averaged across multiple repetitions which provided a single discrete value per kinematic feature, task, and participant. This results in loss of potentially useful information within each repetition as well as more broadly looking at the values of features across multiple repetitions. For example, useful information

regarding fatigue and variability clinical scores can be obtained by looking at the changes of the kinematic measures across multiple repetitions in tasks such as sentence repetition. Future work may explore using time series data of kinematic features rather than single values to potentially capture more clinically relevant features. For example, in a study using an LSTM architecture on the time series of facial landmarks data obtained from participants expressing specific emotions in front of camera was used to distinguish between individuals with Parkinson disease and healthy controls [100]. In this study LSTM showed promising results in analyzing sequential video data to guide clinicians with diagnosis of neurological diseases. The main limitation of this study was small sample size [100]. In the future, as more clinical data becomes available, such architectures can be used to derive clinically meaningful features that are sensitive to temporal changes within multiple repetitions.

Multimodal data

This work only considered 3D video data (colour and depth). Recent studies have shown that multimodal data (both video and audio) can further improve the predictive power of such algorithms [117], [150]–[153]. For example, Neumann et al. demonstrated that combining acoustic signals obtained from the audio signal (i.e. features obtained from acoustic analysis) together with facial kinematic features results in increase predictive power of these such algorithms [117]. Using a set of features extracted from video as well as audio signals, the trained algorithms showed promising results in differentiating between healthy controls, pre-symptomatic ALS, and bulbar ALS, thus, demonstrating the utility of incorporating various speech and facial metrics to assist early diagnosis and monitoring of ALS [117]. Similarly, the future work of this research should consider extracting clinically valid speech features from audio recordings of the speech tasks to potentially increase the predictive power of the regression models.

Collecting Longitudinal Data

These technologies have a great potential for diagnosis and prognosis of neurological disorders as larger volumes of longitudinal data, accounting for objective inter and intra subject variability, is being gathered [154]. Current clinical practice suffers from one-shot assessments, where patients see clinicians once every few months or even years [149]. The performance of the patient may not provide a true picture of their performance due to the day-to-day variability of the

disease characteristics [154]. This is of particular importance in neurological disorders in which the disease representation itself is heterogenous and patient's clinical characteristics may vary from day-to-day [139], [154]. Therefore, collecting repeated assessments reduce intra-participant's variability while increasing the power of the study [154]. In addition, longitudinal study design can be more efficient, cost effective, and result in more robust model selection process [155]. Previously longitudinal orofacial and speech home-based assessments in neurological disorders resulted in characteristics of disease progression and/or response to treatment interventions [96]. The future studies will focus on collecting data longitudinally and modelling progression in addition to symptoms' severity.

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Appendices

A.1 Demographic of Participants for Analytical Validation

Table A-1 Demographic information of participants

Participant Number	Age	Gender	Ethnicity
Participant 1	27	F	Middle eastern
Participant 2	29	F	Middle eastern
Participant 3	34	F	Southeast Asian

A.2 Demographic of Participants for Clinical Validation

Table A-2 Demographic information of participants in MND group

Participant	Diagnosis	Gender	Age	Intelligibility	Speaking Rate	Site Onset	FRS-Bulbar
Participant 1	ALS	M	58	100	150	lower limbs	9
Participant 2	ALS	M	68	44.55	79.9	Bulbar	7
Participant 3	ALS	M	55	100	132.26	lower limbs	11
Participant 4	ALS	M	68	69.09	54.1	Bulbar	7
Participant 5	ALS	F	64	96.36	122.68	upper/lower limb	12
Participant 6	ALS	F	61	90	91.16	lower limbs	10
Participant 7	ALS	M	55	91.82	143.17	upper limbs	11
Participant 8	ALS	F	45	28.18	63.28	lower limbs	7
Participant 9	ALS	M	68	87.27	88.61	upper limbs	12
Participant 10	ALS	F	59	99.09	134.42	lower limbs	9
Participant 11	ALS	F	61	data not collected	data not collected	lower limbs	6
Participant 12	ALS	M	71	92.37	109.09	Bulbar	11
Participant 13	ALS	F	75	98.18	147.32	lower limbs	12
Participant 14	ALS	M	58	92.73	145.7	Bulbar	7
Participant 15	ALS	F	69	100	144.1	lower limbs	12
Participant 16	ALS	F	52	99.09	108.55	lower limbs	11
Participant 17	Kennedy	M	57	data not collected	data not collected	upper limbs	NA
Participant 18	Kennedy	M		97.27	245.35	lower limbs	NA
Participant 19	PLS	F	73	data not collected	data not collected	bulbar	NA

Table A-3 Demographic information of participants in PS group

Participant	**Gender**	**Age**	**Onset Location**	**Months from Stroke**
Participant 1	M	63	1st L.Corona Radiata, R.Pons, L.Frontal 2nd R.Cerebellar	35
Participant 2	M	89	Small right pareital centrum semiovalve old lacunar	18
Participant 3	M	60	NA	4
Participant 4	F	61	Right Caudate	18
Participant 5	M	52	R. thalamic stroke	15
Participant 6	M	66	1) R.frontal subcortical 2)L.thalamic& R.Cerebellat	14
Participant 7	F	81	Frontotemporietal subdural hemorrag	29
Participant 8	M	65	Right ICA/MCA ischemic stroke	40
Participant 9	F	76	Left precentral gyrus	NA
Participant 10	F	81	Right occipital	61
Participant 11	M	71	1st Tight ICA territory watershed infarcts 2nd L. AICA	12
Participant 12	M	67	Cardioembollic	41
Participant 13	M	74	PICA territories bilaterally	13
Participant 14	M	67	Left insular cortex	74
Participant 15	F	76	Left middle cerebral artery	109
Participant 16	F	61	Right posterior cerebral artery	1
Participant 17	M	64	Right middle & posterior cerebral artery	1
Participant 18	M	67	Left middle cerebral artery	1
Participant 19	M	64	Bilateral pontine & occipital	5
Participant 20	F	70	Left fronto-parietal	1
Participant 21	M	21	NA	NA
Participant 22	F	50	Right middle cerebral arthery	2
Participant 23	M	62	Left middle cerebral artery	2
Participant 24	M	72	Right brainstem	1
Participant 25	F	70	Basal Ganglia	0
Participant 26	M	63	1st L.Corona Radiata, R.Pons, L.Frontal 2nd R.Cerebellar	35